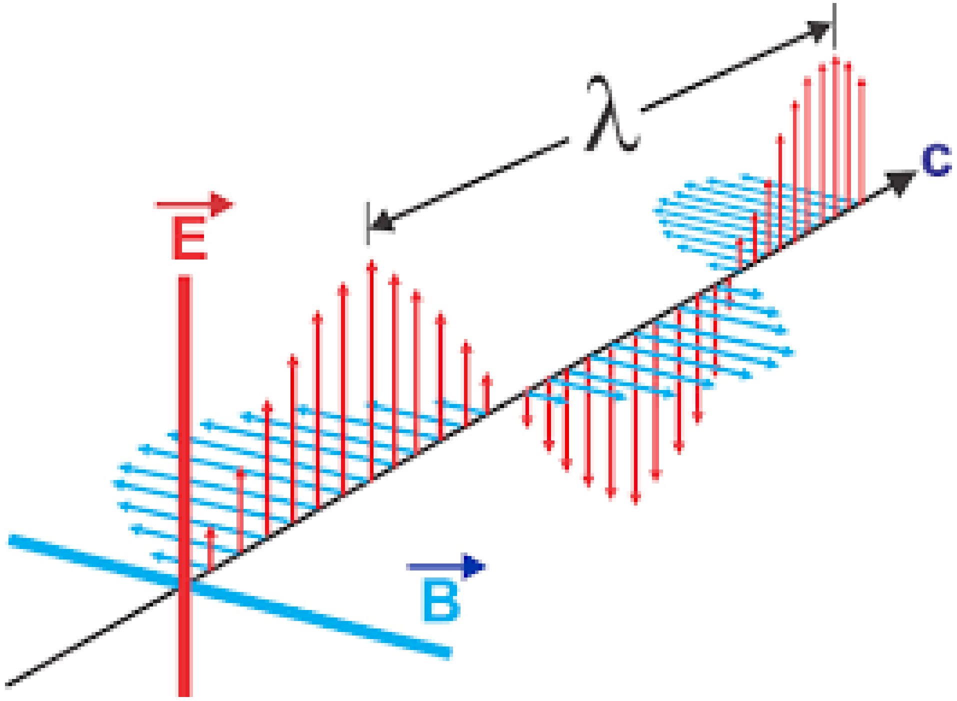
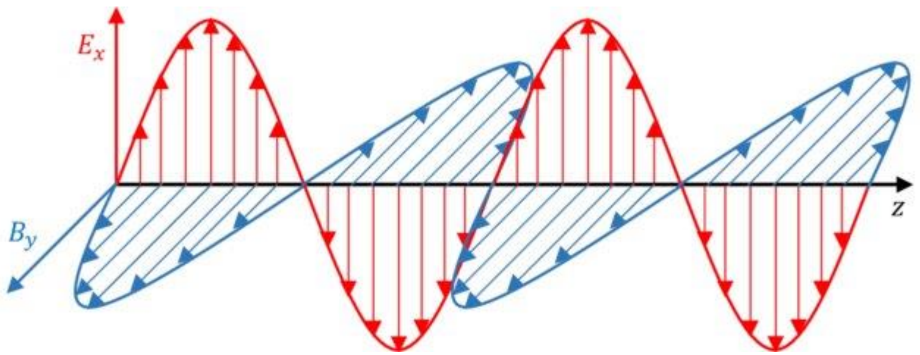
Electromagnetic Wave Propagation & Wave Nature of Light

Lecture-1





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Electromagnetic Wave (or Light) Propagation in Different Mediums

1. In vacuum
2. In matter
3. In conductor
4. Waveguide
5. EM wave interactions in different mediums

Electromagnetic waves in vacuum

Maxwell's equations (wave equation for $\vec{E}$ and $\vec{B}$)

$\vec{\nabla}.\vec{D} = \rho_f$		$\vec{D} = \varepsilon\vec{E}$
$\vec{\nabla}.\vec{B} = 0$		$\vec{B} = \mu\vec{H}$
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	.... (A)	
$\vec{\nabla} \times \vec{B} = \mu\left(\vec{J_f} + \frac{\partial \vec{D}}{\partial t}\right)$		

where $\vec{D} = \varepsilon_0\vec{E} + \vec{P}$ and $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$

About Maxwell's Equations

"Simple enough to imprint on a T-shirt, and yet reach enough to provide new insights throughout a lifetime of study"

—— J. R. Whinnery

"The Teaching of Electromagnetics," *IEEE Trans. Education*, Vol. 33, pp. 3-7 (1990)

For free space,

The charge density ρ_f=0

And the current density $\vec{J_f} = 0$

Also, $\epsilon_r = 1$ *and* $\mu_r = 1$

Hence, $\epsilon = \epsilon_0$ *and* $\mu = \mu_0$

In view of above conditions, the Maxwell's equations;

$\vec{\nabla}.\vec{E} = 0$	**.......................... (a)**	
$\vec{\nabla}.\vec{B} = 0$	..........................(b)	
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	.... (c)	.... (1)
$\vec{\nabla} \times \vec{B} = \mu_0\epsilon_0\frac{\partial \vec{E}}{\partial t}$	.... (d)	

Taking curl of eq 1 (c), we get

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\vec{\nabla} \times \frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B})$$

$$(\vec{\nabla}.\vec{E})\vec{\nabla} - (\vec{\nabla}.\vec{\nabla})\vec{E} = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B})$$

But as $\vec{\nabla}.\vec{E} = 0$, We get

$$\nabla^2 \vec{E} = \frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B})$$

Now using equation 1 (d), we have

$$\nabla^2 \vec{E} = \frac{\partial}{\partial t}\left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}\right)$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Similarly, we will have

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \ldots\ldots\ldots\ldots\ldots\ldots (3$$

Equation 2 and 3 represents the relation between the space and time variation of $\vec{E}$and $\vec{B}$ are called the three-dimensional wave equations for $\vec{E}$and $\vec{B}$

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla(\cancel{\nabla \cdot \mathbf{B}}) - \nabla^2 \mathbf{B} = \nabla \times \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right) = \mu_0 \epsilon_0 \frac{\partial}{\partial t}(\nabla \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

$$\Rightarrow \quad \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

Now standard wave equation is

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2} \dots\dots\dots\dots(4)$$

Comparing eqn 2/3 with 4, we find that $\mu_0\epsilon_0$ has the same magnitude as $\frac{1}{v^2}$

i.e. we have

$$\frac{1}{v^2} = \mu_0\epsilon_0$$

$$v = \frac{1}{\sqrt{\mu_0\epsilon_0}} \dots\dots\dots\dots(5)$$

This is the velocity with which electromagnetic waves travels in free space.

$$\mu_0 = 4\pi \times 10^{-7} Weber/A.m$$

$$\epsilon_0 = 8.85 \times 10^{-12} C^2/N.m^2$$

$$v = \frac{1}{\sqrt{\mu_0\epsilon_0}} = \frac{\frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}} m}{s}$$

$$= 0.02999 \times 10^{10} m/s$$

$$= 2.999 \times 10^{8} m/s$$

$$\cong 3 \times \frac{10^8 m}{s} = c$$

$$\boldsymbol{c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \dots\dots\dots\dots(6)}$$

$\Rightarrow$ in free space e.m. waves travel with speed of light.

Notice the crucial role played by Maxwell's contribution to Ampere's law $(\mu_0\epsilon_0\partial\mathbf{E}/\partial t)$;
without it, the wave equation would not emerge,
and there would be no electromagnetic theory of light.

Electromagnetic wave in matter

Maxwell's equation in matter

$$\left.\begin{aligned}
&\vec{\nabla}\,.\,\vec{\mathrm{D}} = \rho_f \\
&\vec{\nabla}.\,\vec{\mathrm{B}}=0 \\
&\vec{\nabla}\times\vec{\mathrm{E}} = -\frac{\partial\vec{\mathrm{B}}}{\partial t} \\
&\vec{\nabla}\times\vec{\mathrm{H}} = \overrightarrow{J_f} + \frac{\partial\vec{\mathrm{D}}}{\partial t}
\end{aligned}\right\} \quad \text{............ (1)}$$

Consider electromagnetic wave inside linear matter, where there are no free charges or free currents

(Linear medium is an insulator or a non-conductor)

$$\left.\begin{aligned}
&\overrightarrow{J_f} = 0 \\
&\sigma_f = 0
\end{aligned}\right\} \quad \text{............ (2)}$$

For this case, equation (1) becomes

$$\left.\begin{aligned}
&\vec{\nabla}\,.\,\vec{\mathrm{D}} = 0 \\
&\vec{\nabla}.\,\vec{\mathrm{B}}=0 \\
&\vec{\nabla}\times\vec{\mathrm{E}} = -\frac{\partial\vec{\mathrm{B}}}{\partial t} \\
&\vec{\nabla}\times\vec{\mathrm{H}} = \frac{\partial\vec{\mathrm{D}}}{\partial t}
\end{aligned}\right\} \quad \text{............ (3)}$$

Now, for linear, homogeneous and isotropic medium,

$\vec{D} = \epsilon \vec{E}$ and $\vec{H} = \frac{\vec{B}}{\mu}$ .... (4)

Therefore, Maxwell's equation for E and B fields inside a linear, homogeneous and isotropic non- conducting medium become:

$$\vec{\nabla}.\vec{E}=0$$

$$\vec{\nabla}.\vec{B}=0$$

$$\vec{\nabla} \times \vec{E}=-\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B}=\mu\epsilon\frac{\partial \vec{E}}{\partial t} \quad \text{.................... (5)}$$

These relations are almost identical to those for electromagnetic waves in free space

- Linear: The electric and magnetic responses of the material are directly proportional to the electric and magnetic fields applied to it.
- Homogeneous: The material has uniform properties throughout its volume.
- Isotropic: The material has similar properties in any direction of travel for waves through that medium

In vacuum, $c=\frac{1}{\sqrt{\mu o \epsilon o}}$ = 3X 10^8 m/s

But in linear medium, $v=\frac{1}{\sqrt{\mu\epsilon}}$

Which differ from the free space case only with the term $\mu\varepsilon$ instead of $\mu_o\varepsilon_0$.

The velocity of the electromagnetic wave in the material medium;

$$v = \frac{1}{\sqrt{\mu\varepsilon}}$$

From optics, the refractive index

$$n = \frac{speed\ of\ light}{speed\ in\ the\ material\ medium} = \frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_o\varepsilon_o}} = \sqrt{\frac{\mu_r\,\mu_o\varepsilon_r\varepsilon_o}{\mu_o\varepsilon_o}}$$

$$n = \sqrt{\mu_r\varepsilon_r}$$

The solution of the wave equation;

E(r,t) = **E$_0$** $e^{i(\boldsymbol{k}.\boldsymbol{r}-\omega t)}$

and **B**(r,t) = **B$_0$** $e^{i(\boldsymbol{k}.\boldsymbol{r}-\omega t)}$

Where k is the propagation vector and r is the position vector.

Plane wave propagation in free space (E , B and n relationship)

The electric and magnetic field components of the plane EM wave are given by

$$\nabla^2 E = \mu_0 \varepsilon_0 \, \partial^2 E/\partial t^2 \quad \text{..................} \ (1)$$

and $$\nabla^2 B = \mu_0 \varepsilon_0 \, \partial^2 B/\partial t^2 \quad \text{......................} \ (2)$$

Or $$\nabla^2 E - \frac{1}{v^2} \, \partial^2 E/\partial t^2 = 0 \quad \text{........................} \ (3)$$

$$\nabla^2 B - \frac{1}{v^2} \partial^2 B/\partial t^2 = 0 \quad \text{......................} \ (4)$$

Since E and B are function of space and time, their solution can also be assumed as

$$E(r, t) = E(r) \, e^{-i\omega t} \quad \text{..................} \ (5)$$

$$B(r, t) = B(r) \, e^{-i\omega t} \quad \text{....................} \ (6)$$

Substituting these values in equation (3) and (4), we will get

$$e^{-i\omega t}\nabla^2 \vec{E}(r) + \frac{\omega^2}{v^2} \vec{E}(r)e^{-i\omega t} = 0$$

$$\text{or} \quad \nabla^2 \vec{E}(r) + \frac{\omega^2}{v^2} \vec{E}(r) = 0 \quad \ldots\ldots(7)$$

$$\text{and} \quad \nabla^2 \vec{B}(r) + \frac{\omega^2}{v^2} \vec{B}(r) = 0 \quad \ldots\ldots\ldots(8)$$

Since ω is angular frequency

$$\frac{\omega}{v} = \frac{2\pi\upsilon}{v} = \frac{2\pi}{\lambda} = k$$

Where k is the propagation vector

$$\nabla^2 \vec{E}(r) + k^2 \vec{E}(r) = 0 \quad \ldots\ldots\ldots(9)$$

$$\nabla^2 \vec{B}(r) + k^2 \vec{B}(r) = 0 \quad \ldots\ldots\ldots(10)$$

These equations and are of type

$$\nabla^2 \varphi\,(r)\; + k^2\, \varphi(r) = 0 \quad \text{............(11)}$$

known as Helmholtz equation and φ(r) represent one of the six field components .

The electric and magnetic vector obey identical wave equation Viz - Helmholtz equation.

The solution of equation (11) can be written as

$$\varphi(\,x,y,z) = X(\,x)\,Y\,(y)\,Z(z)$$

Substituting these values in equation (11), we will get

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) XYZ + k^2\, XYZ = 0$$

$$\left(YZ\,\frac{\partial^2 X}{\partial x^2} + XZ\,\frac{\partial^2 Y}{\partial y^2} + XY\,\frac{\partial Z^2}{\partial z^2}\right) + k^2\, XYZ = 0$$

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{y}\frac{\partial^2 Y}{\partial y^2} + \frac{1}{z}\frac{\partial Z^2}{\partial z^2} = -\,k^2 \qquad \text{........(12)}$$

Each term of the left is a constant from the variable separation such that

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = - k_x^2$$

$$\frac{1}{y}\frac{\partial^2 Y}{\partial y^2} = - k_y^2$$

$$\frac{1}{z}\frac{\partial^2 Z}{\partial z^2} = - k_z^2$$

Solution of these equations are

X = $X_0 e^{\pm i k_x x}$

Y = $Y_0 e^{\pm i k_y y}$

Z = $Z_0 e^{\pm i k_z z}$

So that $\varphi = X_0 \, Y_0 Z_0 e^{\pm i(k_x x + k_y y + k_z z)}$

Hence solution in terms of E field will be

$E_x = E_{0x} \, e^{\pm i\vec{k}.\vec{r}}$, $E_y = E_{0y} \, e^{\pm i\vec{k}.\vec{r}}$, $E_z = E_{0z} \, e^{\pm i\vec{k}.\vec{r}}$ .. (13)

Same equation for B_x, B_y, B_z

Thus in vector form

$$\vec{E}(\text{ r }) = \widetilde{E_0}\ e^{\pm i\vec{k}.\vec{r}}$$

$$\vec{B}(\text{ r }) = \widetilde{B_0}\ e^{\pm i\vec{k}.\vec{r}}$$

Hence

$$\boldsymbol{\vec{E}(\vec{r}\ ,\text{ t }) = \widetilde{E_0}\ \ e^{-i(\omega t\ \mp\vec{k}.\vec{r})}}$$

and $\boldsymbol{\vec{B}(\ \vec{r}.\ t) = \widetilde{B_0}\ \ e^{-i(\omega t\mp\vec{k}.\vec{r})}}$

Since $\vec{k}\ = \frac{2\pi}{\lambda} = \frac{2\pi\upsilon}{v} = \frac{\omega}{v}\ \hat{n}$

Hence $\vec{E}\ (\ \vec{r}\ ,\text{ t }) = \ \widetilde{E_0}\ \ e^{-i\omega(t\mp\ \frac{\hat{n}.\vec{r})}{v}}$ ………………………(14)

and $\vec{B}\ (\ \vec{r}\ ,\text{ t }) = \widetilde{B_0}\ \ e^{-i\omega(t\mp\frac{\hat{n}.\vec{r})}{v}}$

Where is $\hat{n}$ the unit vector in the propagation direction .

We see that

E_x (r,t) $=E_{0x}$f (t - $\frac{\vec{r}.\hat{n}}{v}$)

E_y (r,t) = E_{0y} f (t - $\frac{\vec{r}.\hat{n}}{v}$)......................(15)

And E_z (r,t) = E_{0z} f (t - $\frac{\vec{r}.\hat{n}}{v}$)

Now $\frac{\partial E_x}{\partial x}$ = E_{0x} f' (t - $\frac{\vec{r}.\hat{n}}{v}$) $\left(\frac{-n_x}{v}\right)$

Where n_x is the x - components of $\vec{n}$

$\frac{\partial E_x}{\partial x}$ = $\left(\frac{-n_x}{v}\right)$ E_{0x} f' (t - $\frac{\vec{r}.\hat{n}}{v}$)

Since $\vec{\nabla}.\vec{E} = 0$ in free space

- $\frac{1}{v}$ f'` (t - $\frac{\vec{r}.\hat{n}}{v}$) $\left\{E_{0x}n_x + E_{0y}n_y + E_{0z}n_z\right\}$ = 0

- $\frac{\widetilde{E_0}.\hat{n}}{v}$ f' (t - $\frac{\vec{r}.\hat{n}}{v}$) = 0

Hence $\widetilde{E_0}.\hat{n}$=0

It means that $\widetilde{E_0}$ is normal to the direction of propagation .

In the same way,

$\widetilde{B_0} \cdot \hat{n} = 0$ ....(17)

Thus B_0 is normal to the direction of propagation.

Let us also find relation between $\widetilde{E_0}$ and $\widetilde{B_0}$

Since $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$(\vec{\nabla} \times \vec{E})x = -\frac{\partial B_x}{\partial t}$

$\frac{\partial E_Z}{\partial y} - \frac{\partial E_y}{\partial z} = -\frac{\partial B_x}{\partial t} = -\frac{\partial}{\partial t}\left\{B_{0x}\, \mathrm{f}\left(\mathrm{t} - \frac{\vec{r}\,.\,\hat{n}}{v}\right)\right\}$ ....(18)

$\frac{\partial E_Z}{\partial y} = \frac{\partial}{\partial y}\left\{E_{0z}\, \mathrm{f}\left(\mathrm{t} - \frac{\vec{r}\,.\,\hat{n}}{v}\right)\right\}$

$= -\frac{n_y}{v}\left\{E_{0z}\; f'\left(\mathrm{t} - \frac{\vec{r}\,.\,\hat{n}}{v}\right)\right\}$

$\frac{\partial E_y}{\partial z} = -\frac{n_z}{v}\left\{E_{0y}\; \mathrm{f}'\left(\mathrm{t} - \frac{\vec{r}\,.\,\hat{n}}{v}\right)\right\}$

Hence

$$(\vec{\nabla} \times \vec{E})x = \left(\left(\frac{-n_y}{v}\right) E_{0z} + \left(\frac{n_z}{v}\right) E_{0y}\right) \left\{f'\left(t - \frac{\vec{r}\,.\,\hat{n}}{v}\right)\right\}$$

$$= \left\{-B_{0x}\ f'\left(t - \frac{\vec{r}\,.\,\hat{n}}{v}\right)\right\}$$

Or $\frac{n_y}{v} E_{0z} - \frac{n_z E_{0y}}{v} = B_{0x}$

Or $n_y E_{0z} - n_z E_{0y} = vB_{0x}$

Similarly, $n_z E_{0x} - n_x E_{0z} = vB_{0y}$

$n_x E_{0y} - n_z E_{0y} = vB_{0z}$

Thus combining three eqs. we will get

$\hat{n} \times \widetilde{E_0} = \text{v}\ \widetilde{B_0}$

Hence $\widetilde{B_0}$ is right angle to both $\widetilde{E_0}$ and $\hat{n}$

Thus $\vec{E}$ an$\vec{B}$ are mutually perpendicular and transverse to the direction of propagation.

$$\vec{E}(\vec{r}\ ,\text{t}) = \widetilde{E_0}\ e^{i(\vec{k}.\vec{r}-\omega t)}$$

$$\vec{B}(\vec{r}.t) = \widetilde{B_0}\ e^{i(\vec{k}.\vec{r}-\omega t)}$$

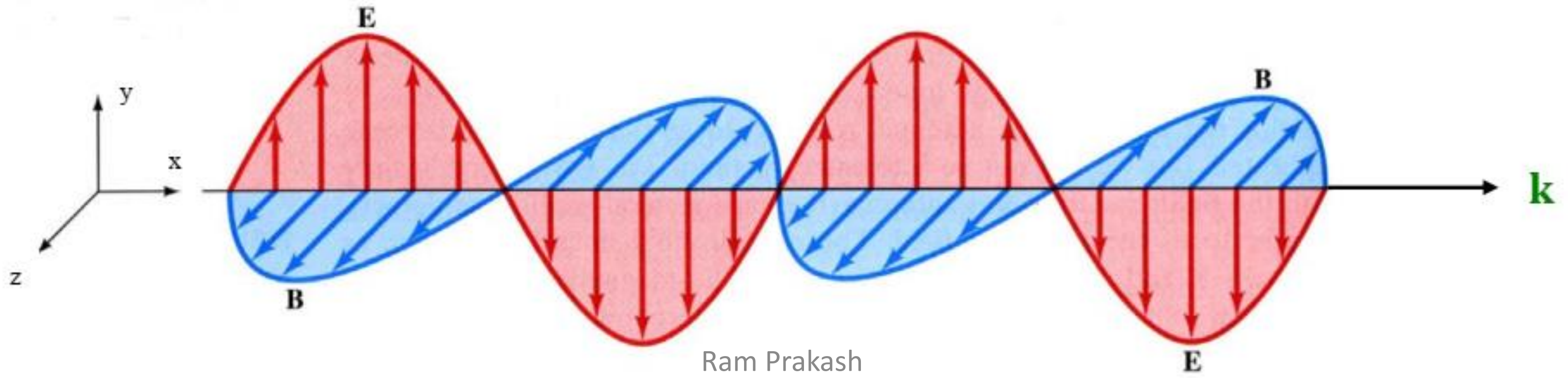
$\frac{\partial}{\partial t} = -i\omega$

and $\frac{\partial}{\partial x} = i k_x$ or $\vec{\nabla} = i\vec{K}$ (in case of 3-D)

Algebraic form of Maxwell's Equations in free space

- $i\,\vec{k}\,.\,\vec{E} = 0$ (i.e. $\vec{E}$ is perpendicular to $\vec{k}$)
- $i\,\vec{k}\,.\,\vec{B} = 0$ (i.e. $\vec{B}$ is perpendicular to $\vec{k}$)
- $i\,\vec{k} \times \vec{E} = i\omega\,\vec{B}$
- $i\,\vec{k} \times \vec{B} = -\,i\varepsilon_o\mu_o\,\omega\,\vec{E}$

E and B are mutually perpendicular to each other,
E and B are perpendicular to the direction of propagation of wave.

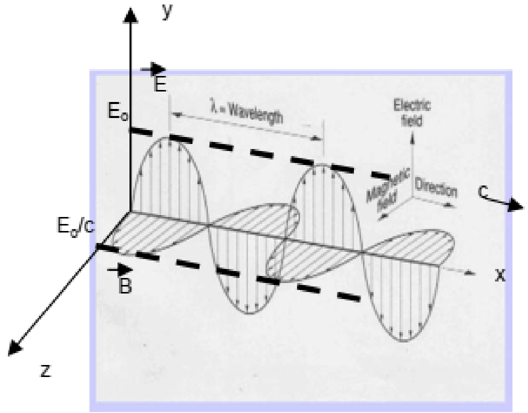


What is the relation between E and B ?
Or can we show that E and B are in same phase at any time in space.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \Rightarrow \quad \mathbf{k} \times \mathbf{E} = \omega \mathbf{B}$$

$$\text{when} \quad \left.\begin{matrix} E \to E_y \\ B \to B_z \\ k \to x \end{matrix}\right\} \Rightarrow \left.\begin{matrix} k E_y = \omega B_z \\ E_y = \dfrac{\omega}{k} B_z \end{matrix}\right\} \Rightarrow \begin{matrix} E_y = cB_z \\ B_z = \dfrac{E_y}{c} \end{matrix}$$

$$\text{But} \quad \left.\begin{matrix} E_y = E_o e^{i(kx - wt)} \\ B_z = B_o e^{i(kx - wt)} \end{matrix}\right\} \Rightarrow \boxed{B_o = \frac{E_o}{c}}$$



Since k/ω is a real number, the electric and magnetic vectors are in phase; thus if at any instant, E is zero then B is also zero, when E attains its maximum value, B also attains its maximum value, etc.

Both E_y and B_z are in same phase.

Summary of Important Properties of Electromagnetic Waves

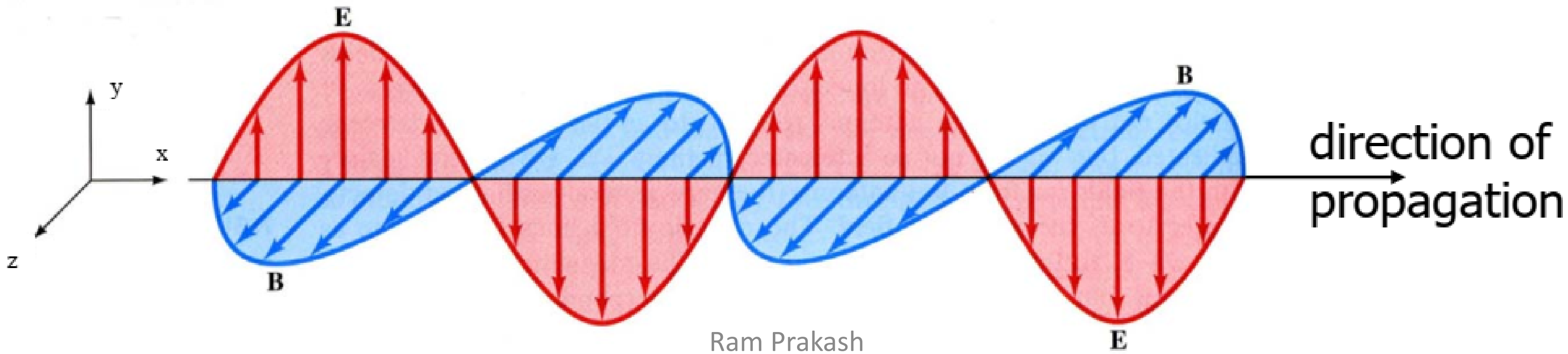
- The solutions (plane wave) of Maxwell's equations are wave-like with both E and B satisfying a wave equation.

$$E_y = E_0 \text{ Sin } (kx - \omega t)$$

$$B_z = B_0 \text{ Sin } (kx - \omega t)$$

- Electromagnetic waves travel through empty space with the speed of light $c = 1/(\mu_0 \varepsilon_0)^{1/2}$.

- The plane wave as represented by above is said to be linearly polarized because the electric vector is always along y-axis and, similarly, the magnetic vector is always along z-axis.

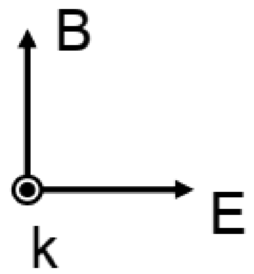


→ The components of the electric and magnetic fields of plane EM waves are perpendicular to each other and perpendicular to the direction of wave propagation. The latter property says that EM waves are transverse waves.

→ The magnitudes of E and B in empty space are related by E/B = c.

$$\frac{E_o}{B_o} = \frac{E}{B} = \frac{\omega}{k} = c$$

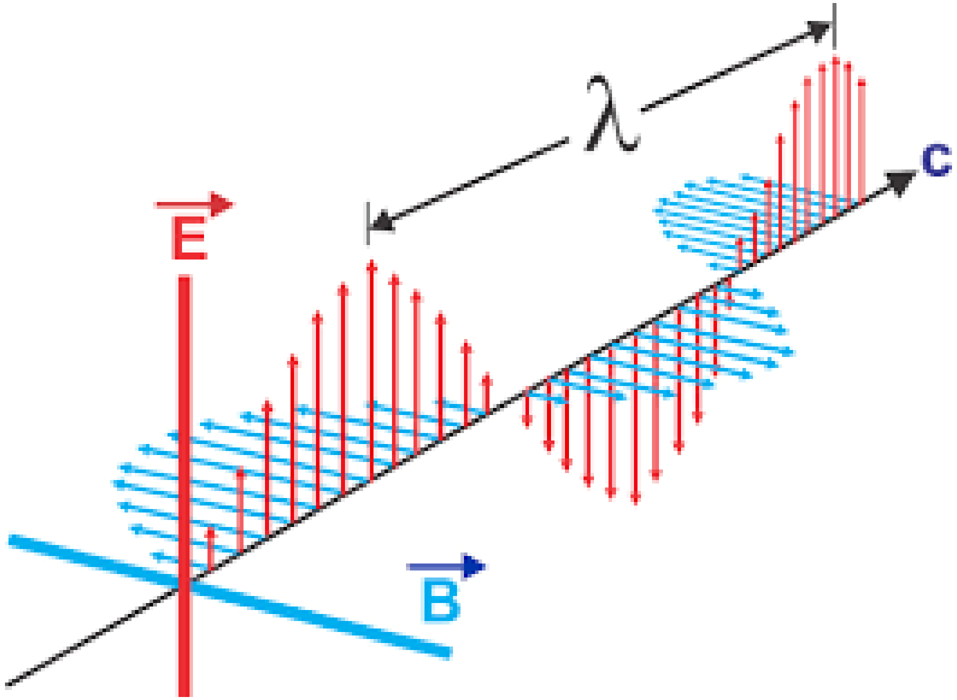
→ Both E and B are at right angles to the direction of propagation. Thus the waves are transverse.

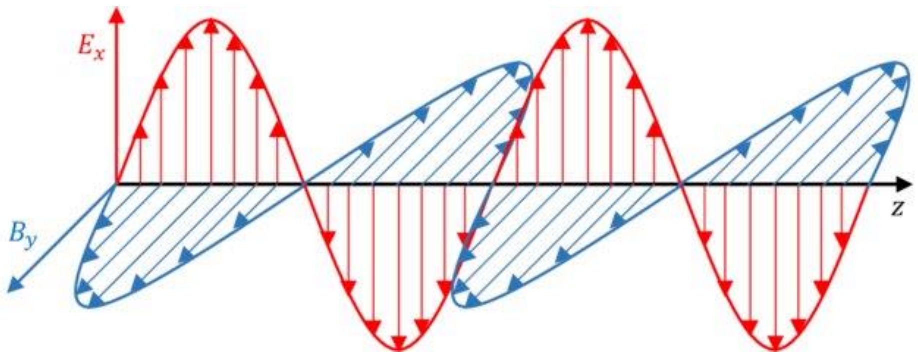


→ The electric and magnetic waves are interdependent; neither can exist without the other. Physically, an electric field varying in time produces a magnetic field varying in space and time; this changing magnetic field produces an electric field varying in space and time and so on. This mutual generation of electric and magnetic fields result in the propagation of the EM waves.

Thanks

Electromagnetic Wave
Part-2





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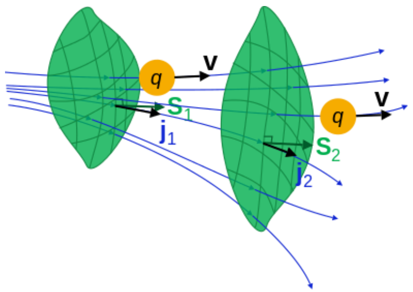
Electromagnetic wave in conductor

Electromagnetic Waves in Conductors



In the case of conductors, $\overrightarrow{J_f}$ is not zero.

The current density in a conductor is proportional to the electric field.

$$\overrightarrow{J_f} = \sigma\vec{E} \quad \dots. (1) \qquad \text{(Ohm's law)}$$

With this, Maxwell's equation for linear conducting media assume the form

$$(i)\ \vec{\nabla}.\vec{E} = \frac{\rho_f}{\epsilon}$$

$$(ii)\ \vec{\nabla}.\vec{B} = 0$$

$$(iii)\ \vec{\nabla} X \vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$(iv)\ \vec{\nabla} X \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial\vec{E}}{\partial t} \quad \dots\dots\dots\dots\dots\dots\dots.. (2)$$

$$\text{(Rate that } q \text{ is flowing through the imaginary surface } S) = \iint_S \mathbf{j}\cdot d\mathbf{S}$$

$$\frac{dq}{dt} + \oiint_S \mathbf{j}\cdot d\mathbf{S} = \Sigma$$

Σ is the net rate that *q* is being generated inside the volume *V* per unit time

The continuity equation for free charge,

$$\vec{\nabla}.\overrightarrow{J_f} = -\frac{\partial \rho_f}{\partial t} \quad \dots\dots\dots\dots\dots\dots\dots\dots.. (3)$$

Using equation (1) in (3), we get

$$\sigma(\vec{\nabla}.\vec{E}\,) = -\frac{\partial \rho_f}{\partial t}$$

But $\vec{\nabla}.\vec{E} = \frac{\rho_f}{\epsilon}$

Therefore, $\sigma(\frac{\rho_f}{\epsilon}) = -\frac{\partial \rho_f}{\partial t}$

Or, $\rho_f(\frac{\sigma}{\epsilon}) + \frac{\partial \rho_f}{\partial t} = 0 \quad \dots\dots\dots\dots\dots. (4)$

So, $\rho_f(t) = \rho_f(0)\, e^{-(\sigma/\epsilon)t} \dots\dots\dots\dots\dots\dots (5)$

$\Rightarrow$ Any initial free charge $\rho_f(0)$ dissipates in a characteristics time $\tau = {}^{\epsilon}/_{\sigma}$.

It means that if we put some free charge on a conductor, it will flow out to the edges.

The time constant τ is a measure of how "good" a conductor is.

- For a good conductor, τ is much less than the other relevant times in the problem (e.g., in oscillatory system, this means $\tau \ll \frac{1}{\omega}$)
- For a "poor" conductor, τ is greater than the characteristics times in the problem ($\tau \gg \frac{1}{\omega}$)

Here we are not interested in this transient behaviour, we will wait for any accumulated free charge to disappear .Then, $\overrightarrow{\rho_f} = 0$.

Then, we will have following Maxwell's equation

$$\left.\begin{aligned} &(i)\vec{\nabla}.\vec{E} = 0 \\ &(ii)\vec{\nabla}.\vec{B}=0 \\ &(iii)\vec{\nabla}X\vec{E}=-\frac{\partial\vec{B}}{\partial t} \\ &(iv)\ \vec{\nabla}X\vec{B}=\mu\sigma\vec{E}+\mu\epsilon\frac{\partial\vec{E}}{\partial t} \end{aligned}\right\} \quad \ldots\ldots\ldots\ldots (6)$$

Now, applying curl to equations (iii) and (iv):

Equation (iii) is:

$$\vec{\nabla}\text{X}(\vec{\nabla}\text{X}\vec{\text{E}}) = -\frac{\partial(\vec{\nabla}\text{X}\vec{\text{B}})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E} = -\frac{\partial}{\partial t}\left(\mu\sigma\vec{E}+\mu\epsilon\frac{\partial\vec{E}}{\partial t}\right)$$

Or, $-\nabla^2\vec{E} = -\mu\epsilon\frac{\partial^2\vec{E}}{\partial t^2} - \mu\sigma\frac{\partial\vec{E}}{\partial t}$

$$\nabla^2\vec{E} = \mu\epsilon\frac{\partial^2\vec{E}}{\partial t^2} + \mu\sigma\frac{\partial\vec{E}}{\partial t} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (7)$$

Equation (iv) is :

$$\vec{\nabla}\text{X}(\vec{\nabla}\text{X}\vec{\text{B}}) = \mu\sigma\overrightarrow{(\vec{\nabla}X\vec{E})}+\mu\epsilon\frac{\partial(\vec{\nabla}X\vec{E})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla}.\vec{B}) - \nabla^2\vec{B} = -\mu\epsilon\frac{\partial^2\vec{B}}{\partial t^2} - \mu\sigma\frac{\partial\vec{B}}{\partial t}$$

$$\nabla^2\vec{B} = \mu\epsilon\frac{\partial^2\vec{B}}{\partial t^2} + \mu\sigma\frac{\partial\vec{B}}{\partial t} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (8)$$

Thus, we will have

$$\boldsymbol{\nabla^2\vec{E} = \mu\epsilon\frac{\partial^2\vec{E}}{\partial t^2} + \mu\sigma\frac{\partial\vec{E}}{\partial t}}$$

$$\boldsymbol{\nabla^2\vec{B} = \mu\epsilon\frac{\partial^2\vec{B}}{\partial t^2} + \mu\sigma\frac{\partial\vec{B}}{\partial t}}$$

Note: 3D wave equations for E and B in a conductor have an additional term that has a single time derivative – which is analogous to a velocity dependent damping term associated with the motion of a 1D mechanical harmonic oscillator.

The general solutions ~in the form of an oscillatory function times a damping term (i.e., a decaying exponential) in the direction of the propagation of the EM wave.

A Complex plane wave type solution for E and B associated with such wave equations are of the general form.

$$\vec{\tilde{E}} = \overrightarrow{\tilde{E}o}\; e^{i(\tilde{K}\, z - \omega\, t)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(9)}$$

$$\vec{\tilde{B}} = \overrightarrow{\tilde{B}o}\; e^{i(\tilde{K}\, z - \quad t)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots.. \text{(10)}$$

With complex wave number

$\tilde{K} = k + iK$ ……………….. (11)

$$\tilde{K} = \tilde{K}(\omega)$$

From Eqs. 9, 10, 7 & 8

$\tilde{K}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega$ ………………….. (12)

Then,

$\tilde{K}^2 = (k + iK)^2 = k^2 - K^2 + 2ikK = \mu\epsilon\omega^2 + i\mu\sigma\omega$

Or $[(k^2 - K^2) - \mu\epsilon\omega^2] + i[2kK - \mu\sigma\omega] = 0$ ……………….. (13)

Only way this relation can be true, when

$(k^2 - K^2) - \mu\epsilon\omega^2 = 0$

$2kK - \mu\sigma\omega = 0$

$(k^2 - K^2) = \mu\epsilon\omega^2$ ………………. (14)

$2kK = \mu\sigma\omega$ …………………….. (15)

We need to solve equations (14) and (15) simultaneously to determine k and Қ.

From equation (15), we get

Қ = $\frac{\mu\sigma\omega}{2k}$ ……………….. (16)

Plugging this into equation (14)

k^2 – $Қ^2$ = k^2 - $(\frac{\mu\sigma\omega}{2k})^2$ = $\mu\epsilon\omega^2$

Multiplying above equation by k^2, we get

k^4 - $(\frac{\mu\sigma\omega}{2})^2 = (\mu\epsilon\omega^2)k^2$

k^4 - $(\mu\epsilon\omega^2)k^2 - \left(\frac{1}{2}\mu\sigma\omega\right)^2 = 0$ ……………… (17)

Let's define x = k^2

a = 1

b= - $(\mu\epsilon\omega^2)$

c= - $(\frac{\mu\sigma\omega}{2})^2$

Then equation (17) becomes

$ax^2+bx+c = 0$

therefore, x= $\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

or, k^2= $\frac{1}{2}$ $[(\mu\epsilon\omega^2 \pm \sqrt{(\mu\epsilon\omega^2)^2+4(\frac{\mu\sigma\omega}{2})^2}$

or, k^2= $\frac{1}{2}(\mu\epsilon\omega^2)[1 \pm \sqrt{1+4\frac{\mu^2\sigma^2\omega^2}{4\mu^2\epsilon^2\omega^4}}$]`

or, k^2= $\frac{1}{2}(\mu\epsilon\omega^2)[1 \pm \sqrt{1+\frac{\sigma^2}{\epsilon^2\omega^2}}$]

or, $k^2 = \frac{1}{2}(\mu\epsilon\omega^2)\left[1 \pm \sqrt{1+(\frac{\sigma}{\epsilon\omega})^2}\right]$ ………………. (18)

on physical grounds,($k^2>0$), we must select the +ve sign .

hence,

$$k^2=\frac{1}{2}\left(\mu\epsilon\omega^2\right)\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right] \ldots\ldots\ldots\ldots\ldots (19)$$

$$\text{or } k=\sqrt{k^2}=\omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right]}$$

$$\text{or } k=\omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right]} \ldots\ldots\ldots\ldots (20)$$

having thus solved for k , we can use either of our original two relations to solve for K . for e.g. $k^2 - K^2 = \mu\epsilon\omega^2$,

$$K^2=k^2-\mu\epsilon\omega^2=\frac{1}{2}\left(\mu\epsilon\omega^2\right)\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right]-\mu\epsilon\omega^2$$

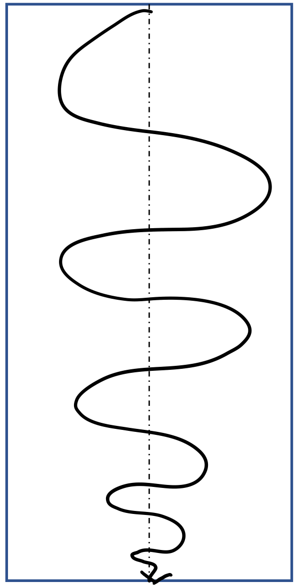
$$\text{Or, } K^2=\frac{1}{2}\left(\mu\epsilon\omega^2\right)\left[-1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right] \ldots\ldots\ldots\ldots\ldots (21)$$

We thus have

$$k=\omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right]}$$

$$K=\omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\left[-1+\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}\right]}$$

The imaginary part of $\widetilde{K}$ results in an attenuation of the wave (decreasing amplitude with increasing z)



$$\vec{\tilde{E}}(z,t) = \vec{\tilde{E}}_0 e^{-Кz}\, e^{i(k\,z-\omega t\,)} \qquad \ldots\ldots\ldots\ldots (23)$$

$$\vec{\tilde{B}}(z,t) = \overrightarrow{\tilde{B}_0} e^{-Кz}\, e^{i(k\,z-\omega t\,)}$$

Skin depth: The distance it takes to reduce the amplitude by a factor of $^1/_e$

$$d = \frac{1}{К} \quad \ldots\ldots\ldots\ldots\ldots (24)$$ where d is the skin depth.

It is a measure of how far the wave penetrates into the conductor. Meanwhile the real part of $\widetilde{K}$ determines the wavelength the propagation speed, and the index of reflection,

$$\lambda = \frac{2\pi}{k} \quad \ldots\ldots\ldots (25), \quad v=\frac{\omega}{k} \quad \ldots\ldots\ldots\ldots (26), \quad n = \frac{ck}{\omega} \quad \ldots\ldots\ldots\ldots (27)$$

Maxwell’s equation (6) impose further constraints fixes relatives amplitudes, phases and polarizations of E and B.

Eq. 6(i) and (ii) rule out any z components: the fields are transverse.

Assuming E is polarized along the x- direction:

$$\vec{\tilde{E}}(z,t) = \tilde{E}o\ e^{-Kz}\ e^{i(k\,z-\omega t)}\ \hat{x} \ \ldots\ldots\ldots\ldots\ldots\ (28)$$

Then equation 6(iii) gives,

$$\vec{\tilde{B}}(z,t) = \frac{\tilde{K}}{\omega}\tilde{E}o\ e^{-Kz}\ e^{i(k\,z-\omega\,t)}\ \hat{y} \ \ldots\ldots\ldots\ (29)$$

Since $B_y = \frac{k}{\omega}E_x$

$\Rightarrow$ E and B Fields are mutually perpendicular.

According to equation (28) and (29), the complex amplitude $\tilde{E}o = Eo\ e^{i\delta_E}$ and $\tilde{B}o = Boe^{i\delta_B}$ (free space case) are related here by

$$Boe^{i\delta_B} = \frac{ke^{i\varphi}}{\omega}Eo\ e^{i\delta_E} \ \ldots\ldots\ (32)$$

where $K = |\tilde{K}| = \sqrt{K^2 + k^2} = \omega\sqrt{\mu\epsilon\sqrt{1+(\frac{\sigma}{\epsilon\omega})^2}} \ \ldots\ldots\ldots\ldots\ (30)$

And $\emptyset = \tan^{-1}\frac{K}{k} \ \ldots\ldots\ldots\ (31)$

Thus, it is evident that the electric and magnetic fields are no longer in phase. In fact,

$\delta_B - \delta_E = \emptyset$ implies that magnetic field lags behind the electric field.

$\tilde{K}$ can be expressed in terms of its modulus and phase:

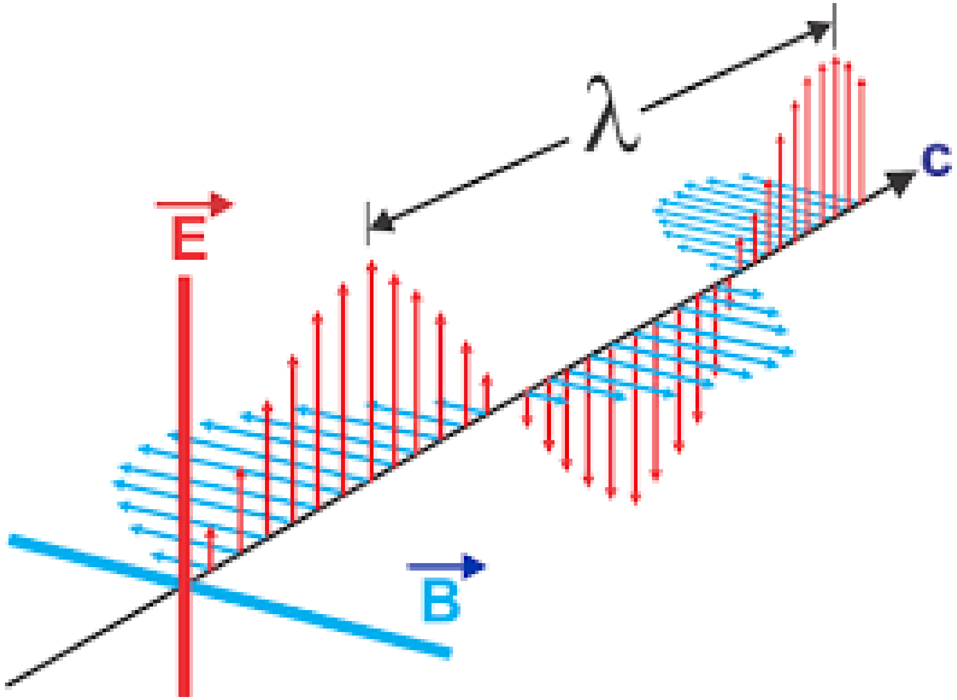
$$\tilde{K} = k\ e^{i\emptyset}$$

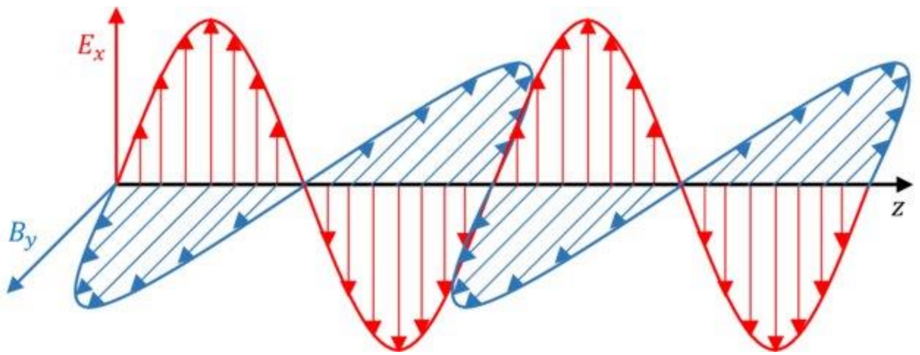
The real amplitudes of $\vec{E}$ and $\vec{B}$ are related by

$$\frac{Eo}{Bo} = \frac{k}{\omega} = \sqrt{\mu\epsilon\sqrt{1+(\frac{\sigma}{\epsilon\omega})^2}} \ \ldots\ldots\ldots\ldots\ldots\ (34)$$

Thanks

EM Wave & Waveguides
Part-3





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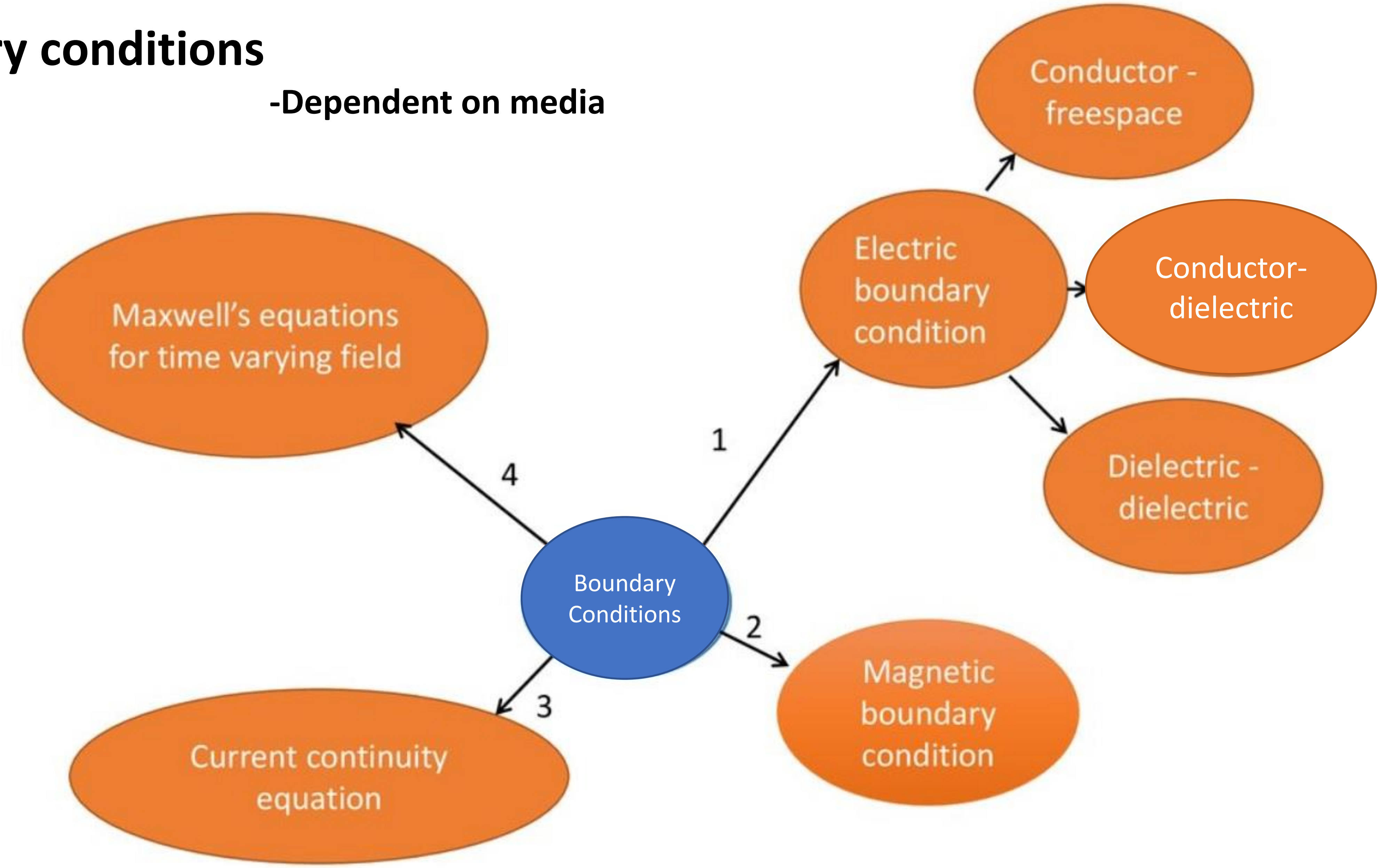
Boundary conditions

-Dependent on media



In general, the fields **E**, **B**, **D**, and **H** will be discontinuous at a boundary between two different media, or at a surface that carries charge density σ or current density **K**.

$$\nabla \cdot \vec{D} = \rho_f \quad \Longrightarrow \quad D_{1n} - D_{2n} = \rho_f$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \Longrightarrow \quad E_{1t} = E_{2t},$$

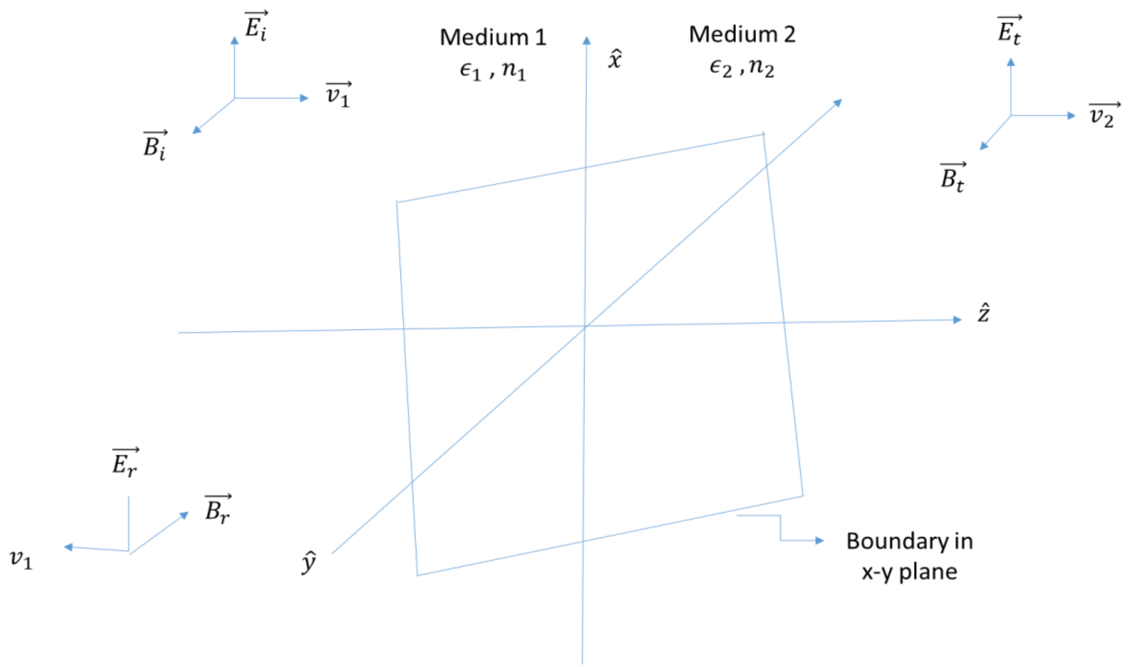
$$\nabla \cdot \vec{B} = 0, \quad \Longrightarrow \quad B_{1n} = B_{2n},$$

$$\nabla \times \vec{H} = \vec{J_f} + \frac{\partial \vec{D}}{\partial t}, \quad \Longrightarrow \quad H_{1t} - H_{2t} = J_f$$

Boundary Conditions

What happens when an electromagnetic wave passes from one linear / homogeneous medium into another (e.g., vacuum, gas, air, water, oil)?

A monochromatic plane electromagnetic wave of frequency ω propagating in $+\hat{z}$ direction, which is linearly polarized in $+\hat{x}$ direction encounters a boundary between two linear media, lies in the x-y plane. Thus, this electromagnetic wave approaches the boundary from left and is at normal incidence to the boundary:



i : incident
r : reflected
t : transmitted

<u>Boundary condition:1</u> The normal component of D is continuous across the interface at z=0:

$$D_1^{\perp} = D_2^{\perp}$$

$\epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp}$ .... (1) (as $\vec{D} = \epsilon \vec{E}$)

<u>Boundary condition:2</u> The tangent component of E is continuous across the interface at z=0:

$E_1^{\parallel} = E_2^{\parallel}$ .... (2)

<u>Boundary condition:3</u> The normal component of B is continuous across the interface at z=0:

$B_1^{\perp} = B_2^{\perp}$ .... (3)

<u>Boundary condition:4</u> The tangential component of H is continuous across the interface at z=0:

$H_1^{\parallel} = H_2^{\parallel}$ .... (4) $\left(\frac{B_1^{\parallel}}{\mu_1} = \frac{B_2^{\parallel}}{\mu_2}\right)$

Medium:1

Incident electromagnetic wave:

Propagates in $+\hat{z}$ – direction (i.e., $\widehat{k_i} = +\widehat{k_1}$) with polarization $+\hat{x}$

Therefore, $\overrightarrow{\widetilde{E}_i}(z,t) = \widetilde{E_{0i}}\, e^{i(k_1 z-\omega t)}\ \hat{x}$ .... (5)

$\overrightarrow{\widetilde{B}_i}(z,t) = \frac{\widehat{k_i} \mathrm{x}\, \overrightarrow{\widetilde{E}_i}(z,t)}{v_1}$

From $\hat{n} \times \widetilde{E_0} = v\, \widetilde{B_0}$ where $\hat{n}$ ~unit vector in the propagation direction

$\overrightarrow{\widetilde{B}_i}(z,t) = \frac{\widetilde{E_{0i}}\, e^{i(k_1 z-\omega t)}}{v_1}\ \hat{y}$ .... (6)

Reflected electromagnetic wave:

$\overrightarrow{\widetilde{E}_r}(z,t) = \widetilde{E_{0r}}\, e^{i(-k_1 z-\omega t)}\ \hat{x}$ .... (7)

$\overrightarrow{\widetilde{B}_r}(z,t) = \frac{\widehat{k_r} X\, \overrightarrow{\widetilde{E}_r}(z,t)}{v_1}$

$\overrightarrow{\widetilde{B}_r}(z,t) = -\frac{\widetilde{E_{0r}}\, e^{i(-k_1 z-\omega t)}}{v_1}\ \hat{y}$ .... (8)

<u>Medium 2:</u>

Transmitted electromagnetic wave:

$$\vec{\tilde{E}}_t(z,t) = \tilde{E}_{0t}\, e^{i(k_2 z-\omega t)}\ \hat{x} \qquad \text{.....................} \ (9)$$

$$\vec{\tilde{B}}_t(z,t) = \frac{\hat{k}_t \times \vec{\tilde{E}}_t(z,t)}{v_2}$$

$$\vec{\tilde{B}}_t(z,t) = \frac{\tilde{E}_{0t}\, e^{i(k_2 z-\omega t)}}{v_2}\ \hat{y} \qquad \text{..........................} \ (10)$$

At the interface, i.e., at z=0 (in the x-y plane) the boundary conditions 1-4 must be satisfied for total E and B Fields present on either side of the interface between the two media:

Boundary condition 1: Normal D continuous at z=0: (refers to the x-y boundary i.e. in the $+\hat{z}$ direction)

$\epsilon_1 E^{\perp}_{1tot}$ (z = 0, t) = $\epsilon_2 E^{\perp}_{2tot}\,(z = 0, t)$ ……………… (11)

Boundary condition 2: Tangential E continuous at z=0: (refers to the x-y boundary i.e. in the x-y plane)

$E^{\parallel}_{1tot}(z = 0, t) = E^{\parallel}_{2tot}$ (z=0, t) ……………. (12)

Boundary condition 3:

$B^{\perp}_{1tot}(z = 0, t) = B^{\perp}_{2tot}\,(z = 0, t)$ …………… (13)

Boundary condition 4: Tangential H is continuous at z=0:

$\frac{B^{\parallel}_{1tot}(z=0,t)}{\mu_1} = \frac{B^{\parallel}_{2tot}(z=0,t)}{\mu_2}$ ……………. (14)

For plane electromagnetic wave at normal incidence on the boundary at z=0 lying in the x-y plane, note that no components of E or B (incident, reflected or transmitted) are allowed to be along $\pm z$ directions because of the E and B fields transversality requirements → arising from the constraints imposed by Maxwell's equations.

Because of this boundary condition 1 & 3 impose no restriction here;

$E_{1tot}^{\perp} = 0;\ E_{2tot}^{\perp}$=0

And $B_{1tot}^{\perp}$=0; $B_{2tot}^{\perp} = 0$

Only restriction on plane electromagnetic wave propagating with normal incidence on the boundary at z= 0 are imposed by boundary condition 2 & 4.

In medium 1 i.e. z≤0:

$$\left.\begin{aligned} \overrightarrow{\widetilde{E_{1tot}^{\parallel}}} &= \overrightarrow{\widetilde{E}_i} + \overrightarrow{\widetilde{E}_r} \\ \frac{\overrightarrow{\widetilde{B_{1tot}^{\parallel}}}}{\mu_1} &= \frac{\overrightarrow{\widetilde{B}_i}}{\mu_1} + \frac{\overrightarrow{\widetilde{B}_r}}{\mu_1} \end{aligned}\right\} \qquad \text{..........} \quad (15)$$

In medium 2 ; z≥0:

$$\overrightarrow{\widetilde{E_{2tot}^{\parallel}}} = \overrightarrow{\widetilde{E}_t} \text{ and } \frac{\overrightarrow{\widetilde{B_{2tot}^{\parallel}}}}{\mu_2} = \frac{\overrightarrow{\widetilde{B}_t}}{\mu_2} \qquad \text{.............} \quad (16)$$

Then from boundary condition 2:

$$\overrightarrow{\widetilde{E_{1tot}^{\parallel}}}\bigg|_{z=0} = \overrightarrow{\widetilde{E_{2tot}^{\parallel}}}\bigg|_{z=0}$$

$\vec{\tilde{E}_i}$ (z=0,t) + $\vec{\tilde{E}_r}$ (z = 0, t) = $\vec{\tilde{E}_t}$ (z=0, t) (from equation 15 & 16)

Or, $\widetilde{E_{0i}}\, e^{-i\omega t}$ + $\widetilde{E_{0r}}\, e^{-i\omega t}$ = $\widetilde{E_{0t}}\, e^{-i\omega t}$ (using equation 5, 7 & 8)

Or , $\widetilde{E_{0i}}$+ $\widetilde{E_{0r}}$= $\widetilde{E_{0t}}$ .... (17)

Using boundary condition 4:

$$\left.\frac{\overrightarrow{\widetilde{B_{1tot}^{\parallel}}}}{\mu_1}\right|_{z=0} = \left.\frac{\overrightarrow{\widetilde{B_{2tot}^{\parallel}}}}{\mu_1}\right|_{z=0}$$

$$\frac{\vec{\tilde{B}_i}\ (z=0,t)}{\mu_1} + \frac{\vec{\tilde{B}_r}\ (z=0,t)}{\mu_1} = \frac{\vec{\tilde{B}_t}\ (z=0,t)}{\mu_2}$$ .... (using equation 15)

$$\frac{\widetilde{E_{0i}}\, e^{-i\omega t}}{\mu_1 v_1} - \frac{\widetilde{E_{0r}}\, e^{-i\omega t}}{\mu_1 v_1} = \frac{\widetilde{E_{0t}}\, e^{-i\omega t}}{\mu_2 v_2}$$ .... (using equation 6,8 & 10)

$$\frac{\widetilde{E_{0i}}}{\mu_1 v_1} - \frac{\widetilde{E_{0r}}}{\mu_1 v_1} = \frac{\widetilde{E_{0t}}}{\mu_2 v_2}$$ .... (18)

Let us define $\beta = \frac{\mu_1 v_1}{\mu_2 v_2}$

Then, equation (18) becomes

$$\boldsymbol{\widetilde{E_{0i}} - \widetilde{E_{0r}} = \beta \widetilde{E_{0t}}}$$ ....(19)

From (17) & (19)

$$2\widetilde{E_{0i}} = (1+\beta)\,\widetilde{E_{0t}}$$

Or, $\widetilde{E_{0t}} = \frac{2}{(1+\beta)}\widetilde{E_{0i}}$ .... (20)

$$2\widetilde{E_{0r}} = (1-\beta)\,\widetilde{E_{0t}}$$

Or, $\widetilde{E_{0t}} = \frac{2}{(1-\beta)}\widetilde{E_{0r}}$ … … … … . (21)

Substituting equation (21) into (20)

$$\widetilde{E_{0r}} = \frac{(1-\beta)}{(1+\beta)}\widetilde{E_{0i}}$$

So, $\frac{\widetilde{E_{0r}}}{\widetilde{E_{0i}}} = \frac{(1-\beta)}{(1+\beta)}$ **and** $\frac{\widetilde{E_{0t}}}{\widetilde{E_{0i}}} = \frac{2}{(1+\beta)}$ .... (22)

If $\mu_1 = \mu_2 \approx \mu_0$, then

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

then $\frac{\widetilde{E_{0r}}}{\widetilde{E_{0i}}} = \frac{v_2 - v_1}{v_2 + v_1} = \frac{n_1 - n_2}{n_1 + n_2}$

and $\frac{\widetilde{E_{0t}}}{\widetilde{E_{0i}}} = = \frac{2v_2}{v_2 + v_1} = \frac{2n_1}{n_1 + n_2}$

$$\boxed{\begin{aligned} R &= \left|\frac{\widetilde{E_{or}}}{\widetilde{E_{oi}}}\right|^2 = \left|\frac{n_1 - n_2}{n_1 + n_2}\right|^2 \\ T &= \left|\frac{\widetilde{E_{ot}}}{\widetilde{E_{oi}}}\right|^2 = \frac{4\,{n_1}^2}{(n_1 + n_2)^2} \end{aligned}}$$

Waveguide

A waveguide is a structure that guides waves, such as electromagnetic waves or sound, with minimal loss of energy by restricting the transmission of energy to one direction.

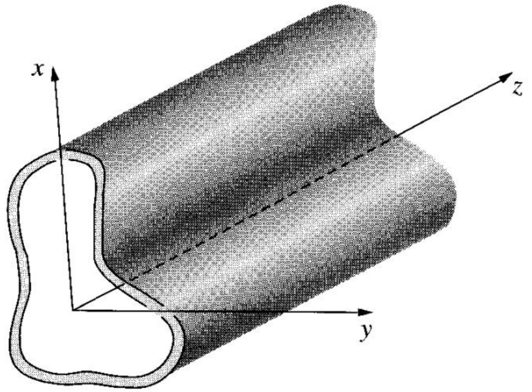
Consider EM waves confined to the interior of a hollow pipe, or waveguide.

Let us assume that the waveguide is a perfect conductor, so that

$$\vec{E} = 0 \; and \; \vec{B} = 0 \text{ (Inside the material itself)}$$

The boundary conditions at the inner wall;

$$E'' = 0, B^{\perp} = 0 \; \ldots \ldots \ldots \ldots \ldots (1)$$



Reflection at a Conducting Surface

We are interested in monochromatic waves that propagate down the tube, so $\vec{E}$ and $\vec{B}$ have the form

Generalized boundary conditions

$$\vec{\tilde{E}}(x, y, z, t) = \vec{\tilde{E}_0}(x, y)e^{i(k_z z - \omega t)}$$

$$\vec{\tilde{B}}(x, y, z, t) = \vec{\tilde{B}_0}(x, y)e^{i(k_z z - \omega t)}$$

.... (2)

(Direction of propagation is along $+\hat{z}$ direction)

In the interior of the waveguide, away from the walls, Maxwell's eq[n]s must be satisfied, which is for empty space or air with $\varepsilon \approx \varepsilon_0$ and $\mu = \mu_0$ are

$\vec{\nabla}.\vec{\tilde{E}} = 0$	$\vec{\nabla}.\vec{\tilde{B}} = 0$	.... (3)
$\vec{\nabla} \times \vec{\tilde{E}} = -\dfrac{\partial \vec{\tilde{B}}}{\partial t}$	$\vec{\nabla} \times \vec{\tilde{B}} = \mu_0 \varepsilon_0 \dfrac{\partial \vec{\tilde{E}}}{\partial t} = \dfrac{1}{c^2}\dfrac{\partial \vec{\tilde{E}}}{\partial t}$	

What restrictions arising from the boundary conditions in (1) are imposed on $\vec{\tilde{E}}$ and $\vec{\tilde{B}}$ in above eq (3) ?

Our $\vec{\tilde{E}}$ and $\vec{\tilde{B}}$ fields interior to the wave guide will be of the forms:

$$\vec{\tilde{E}}(x, y, z, t) = \overrightarrow{\widetilde{E_0}}(x, y) e^{i(k_z z - \omega t)}$$

With $\overrightarrow{\widetilde{\boldsymbol{E_0}}}(\boldsymbol{x}, \boldsymbol{y}) = \widetilde{\boldsymbol{E_{0x}}}(\boldsymbol{x}, \boldsymbol{y})\hat{\boldsymbol{x}} + \widetilde{\boldsymbol{E_{0y}}}(\boldsymbol{x}, \boldsymbol{y})\hat{\boldsymbol{y}} + \widetilde{\boldsymbol{E_{0z}}}(\boldsymbol{x}, \boldsymbol{y})\hat{\boldsymbol{z}} \ldots \ldots \ldots \ldots \ldots \ldots \ldots \ldots . . (\mathbf{4})$

and $\vec{\tilde{B}}(x, y, z, t) = \vec{\tilde{B}}_0(x, y) e^{i(k_z z - \omega t)}$

with $\vec{\tilde{\boldsymbol{B}}}_{\boldsymbol{0}}(\boldsymbol{x}, \boldsymbol{y}) = \widetilde{\boldsymbol{B_{0x}}}(\boldsymbol{x}, \boldsymbol{y})\hat{\boldsymbol{x}} + \widetilde{\boldsymbol{B_{0y}}}(\boldsymbol{x}, \boldsymbol{y})\hat{\boldsymbol{y}} + \widetilde{\boldsymbol{B_{0z}}}(\boldsymbol{x}, \boldsymbol{y})\hat{\boldsymbol{z}} \ldots \ldots \ldots \ldots \ldots \ldots \ldots \ldots . . (\mathbf{5})$

Let us insert these expressions into Faraday's law and Ampere's law in eq^n (3), we get

Faraday's law:

$$\frac{\partial \widetilde{E_{0y}}}{\partial x} - \frac{\partial \widetilde{E_{0x}}}{\partial y} = i\omega \widetilde{B_{0z}} \dots \dots \dots \dots \dots \dots \dots \dots \dots (6)$$

$$\frac{\partial \widetilde{E_{0z}}}{\partial y} - \frac{\partial \widetilde{E_{0y}}}{\partial z} = i\omega \widetilde{B_{0x}} \dots \dots \dots \dots \dots \dots \dots \dots \dots (7)$$

Now $\frac{\partial \widetilde{E_{0y}}}{\partial z} = ik_z\widetilde{E_{0y}}$ → because of z dependence of the factor $e^{i(k_z z-\omega t)}$ so Eq. (7) becomes

$$\frac{\partial \widetilde{E_{0z}}}{\partial y} - ik_z\widetilde{E_{0y}} = i\omega \widetilde{B_{0x}} \dots \dots \dots \dots \dots \dots \dots \dots \dots (8)$$

Finally,

$$\frac{\partial \widetilde{E_{0x}}}{\partial z} - \frac{\partial \widetilde{E_{0z}}}{\partial x} = i\omega \widetilde{B_{0y}} \dots \dots \dots \dots \dots \dots \dots \dots \dots (9)$$

Again as $\frac{\partial \widetilde{E_{0x}}}{\partial z} = ik_z\widetilde{E_{0x}}$, we have

$$ik_z\widetilde{E_{0x}} - \frac{\partial \widetilde{E_{0z}}}{\partial x} = i\omega \widetilde{B_{0y}} \dots \dots \dots \dots \dots \dots \dots \dots . . (10)$$

Thus, we have

$$\boxed{\begin{aligned} &\frac{\partial \widetilde{E_{0y}}}{\partial x} - \frac{\partial \widetilde{E_{0x}}}{\partial y} = i\omega \widetilde{B_{0z}} \\ &\frac{\partial \widetilde{E_{0z}}}{\partial y} - ik_z \widetilde{E_{0y}} = i\omega \widetilde{B_{0x}} \\ &ik_z \widetilde{E_{0x}} - \frac{\partial \widetilde{E_{0z}}}{\partial x} = i\omega \widetilde{B_{0y}} \end{aligned}} \quad \begin{aligned} &\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(a)} \\ &\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(b)} \qquad \ldots\ldots\ldots\ldots\ldots \text{(11)} \\ &\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(c)} \end{aligned}$$

Ampere's law

$$\frac{\partial \widetilde{B_{0y}}}{\partial x} - \frac{\partial \widetilde{B_{0x}}}{\partial y} = \frac{-i\omega}{c^2} \widetilde{E_{0z}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (12)$$

$$\frac{\partial \widetilde{B_{0z}}}{\partial y} - \frac{\partial \widetilde{B_{0y}}}{\partial z} = \frac{-i\omega}{c^2} \widetilde{E_{0x}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (12)$$

But $\frac{\partial \widetilde{B_{0y}}}{\partial z} = ik_z \widetilde{B_{0y}}$

$$\frac{\partial \widetilde{B_{0z}}}{\partial y} - ik_z \widetilde{B_{0y}} = \frac{-i\omega}{c^2} \widetilde{E_{0x}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (14)$$

$$\frac{\partial \widetilde{B_{0x}}}{\partial z} - \frac{\partial \widetilde{B_{0z}}}{\partial x} = \frac{-i\omega}{c^2} \widetilde{E_{0y}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (15), \qquad ik_z \widetilde{B_{0x}} - \frac{\partial \widetilde{B_{0z}}}{\partial x} = \frac{-i\omega}{c^2} \widetilde{E_{0y}} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (16)$$

Thus, we have

$$\boxed{\begin{aligned} &\frac{\partial \widetilde{B_{0y}}}{\partial x} - \frac{\partial \widetilde{B_{0x}}}{\partial y} = \frac{-i\omega}{c^2}\widetilde{E_{0z}} \\ &\frac{\partial \widetilde{B_{0z}}}{\partial y} - ik_z\widetilde{B_{0y}} = \frac{-i\omega}{c^2}\widetilde{E_{0x}} \\ &ik_z\widetilde{B_{0x}} - \frac{\partial \widetilde{B_{0z}}}{\partial x} = \frac{-i\omega}{c^2}\widetilde{E_{0y}} \end{aligned}}
\begin{aligned} &\text{.............................. (a)} \\ &\text{.............................. (b)} \qquad \text{................. (17)} \\ &\text{.............................. (c)} \end{aligned}$$

Using eqn 11 (c) and 17 (b), we get

$$ik_z\widetilde{E_{0x}} - \frac{\partial \widetilde{E_{0z}}}{\partial x} = i\omega\left(\frac{\partial \widetilde{B_{0z}}}{\partial y} + \frac{i\omega}{c^2}\widetilde{E_{0x}}\right)\frac{1}{ik_z}$$

Or

$$ik_z\widetilde{E_{0x}} - \frac{\partial \widetilde{E_{0z}}}{\partial x} = \frac{\omega}{k_z}\frac{\partial \widetilde{B_{0z}}}{\partial y} + \frac{i\omega^2}{k_z c^2}\widetilde{E_{0x}}$$

$$\widetilde{E_{0x}} = \frac{1}{ik_z - \frac{i\omega^2}{k_z c^2}}\left(\frac{\omega}{k_z}\frac{\partial \widetilde{B_{0z}}}{\partial y} + \frac{\partial \widetilde{E_{0z}}}{\partial x}\right)$$

$$\widetilde{E_{0x}} = \frac{1}{\frac{i}{k_z}\left({k_z}^2 - \frac{\omega^2}{c^2}\right)}\left(\frac{\omega}{k_z}\frac{\partial \widetilde{B_{0z}}}{\partial y} + \frac{\partial \widetilde{E_{0z}}}{\partial x}\right)$$

$$\widetilde{E_{0x}} = \frac{ik_z}{\left(\frac{\omega^2}{c^2} - {k_z}^2\right)}\left(\frac{\omega}{k_z}\frac{\partial \widetilde{B_{0z}}}{\partial y} + \frac{\partial \widetilde{E_{0z}}}{\partial x}\right)$$

$$\boldsymbol{\widetilde{E_{0x}} = \frac{i}{\left(\frac{\omega}{c}\right)^2 - {k_z}^2}\left(k_z\frac{\partial \widetilde{E_{0z}}}{\partial x} + \omega\frac{\partial \widetilde{B_{0z}}}{\partial y}\right)} \quad \ldots\ldots\ldots\ldots\ldots\ldots \mathbf{(18)}$$

Similarly, we can get

$$\widetilde{E_{0y}} = \frac{i}{\left(\frac{\omega}{c}\right)^2 - {k_z}^2}\left(k_z\frac{\partial \widetilde{E_{0z}}}{\partial y} - \omega\frac{\partial \widetilde{B_{0z}}}{\partial x}\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots (19)$$

$$\widetilde{B_{0x}} = \frac{i}{\left(\frac{\omega}{c}\right)^2 - {k_z}^2}\left(k_z\frac{\partial \widetilde{B_{0z}}}{\partial x} - \omega\frac{\partial \widetilde{E_{0z}}}{\partial y}\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots (20)$$

$$\widetilde{B_{0y}} = \frac{i}{\left(\frac{\omega}{c}\right)^2 - {k_z}^2}\left(k_z\frac{\partial \widetilde{B_{0z}}}{\partial y} + \omega\frac{\partial \widetilde{E_{0z}}}{\partial x}\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots (21)$$

Now, we need to insert in 18-21 in remaining two Maxwell's equations

$$\frac{\partial \widetilde{E_{0x}}}{\partial x}+\frac{\partial \widetilde{E_{0y}}}{\partial y}+\frac{\partial \widetilde{E_{0z}}}{\partial z}=0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots.(22)$$

$$\frac{\partial \widetilde{B_{0x}}}{\partial x}+\frac{\partial \widetilde{B_{0y}}}{\partial y}+\frac{\partial \widetilde{B_{0z}}}{\partial z}=0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots.(23)$$

Substituting equation 18 and 19 in 22, we get

$$\frac{i}{\left(\frac{\omega}{c}\right)^2-{k_z}^2}\left[k_z\frac{\partial^2\widetilde{E_{0z}}}{\partial x^2}+\omega\frac{\partial^2\widetilde{B_{0z}}}{\partial x\partial y}\right]+\frac{i}{\left(\frac{\omega}{c}\right)^2-{k_z}^2}\left[k_z\frac{\partial^2\widetilde{E_{0z}}}{\partial y^2}-\omega\frac{\partial^2\widetilde{B_{0z}}}{\partial x\partial y}\right]+ik_z\widetilde{E_{0z}}=0$$

$$\frac{ik_z}{\left(\frac{\omega}{c}\right)^2-{k_z}^2}\left(\frac{\partial^2\widetilde{E_{0z}}}{\partial x^2}+\frac{\partial^2\widetilde{E_{0z}}}{\partial y^2}\right)+ik_z\widetilde{E_{0z}}=0$$

$$\frac{\partial^2\widetilde{E_{0z}}}{\partial x^2}+\frac{\partial^2\widetilde{E_{0z}}}{\partial y^2}+ik_z\frac{\left(\frac{\omega}{c}\right)^2-{k_z}^2}{ik_z}\widetilde{E_{0z}}=0$$

$$\frac{\partial^2\widetilde{E_{0z}}}{\partial x^2}+\frac{\partial^2\widetilde{E_{0z}}}{\partial y^2}+\left\{\left(\frac{\omega}{c}\right)^2-{k_z}^2\right\}\widetilde{E_{0z}}=0$$

$$\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\left(\frac{\omega}{c}\right)^2-k_z{}^2\right]\widetilde{E_{0z}}=0 \ldots\ldots\ldots\ldots\ (24)$$

Similarly, we get

$$\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\left(\frac{\omega}{c}\right)^2-k_z{}^2\right]\widetilde{B_{0z}}=0 \ldots\ldots\ldots\ldots\ (25)$$

Case-1 Longitudinal component of $\vec{\tilde{E}} = 0$

$\widetilde{E_{0z}} = 0\ but\ \widetilde{B_{0z}} \neq 0$ →transverse electric (TE) waves

Case 2 Longitudinal component of $\vec{\tilde{B}} = 0$

$\widetilde{B_{0z}} = 0\ but\ \widetilde{E_{0z}} \neq 0$ →transverse magnetic (TM) waves

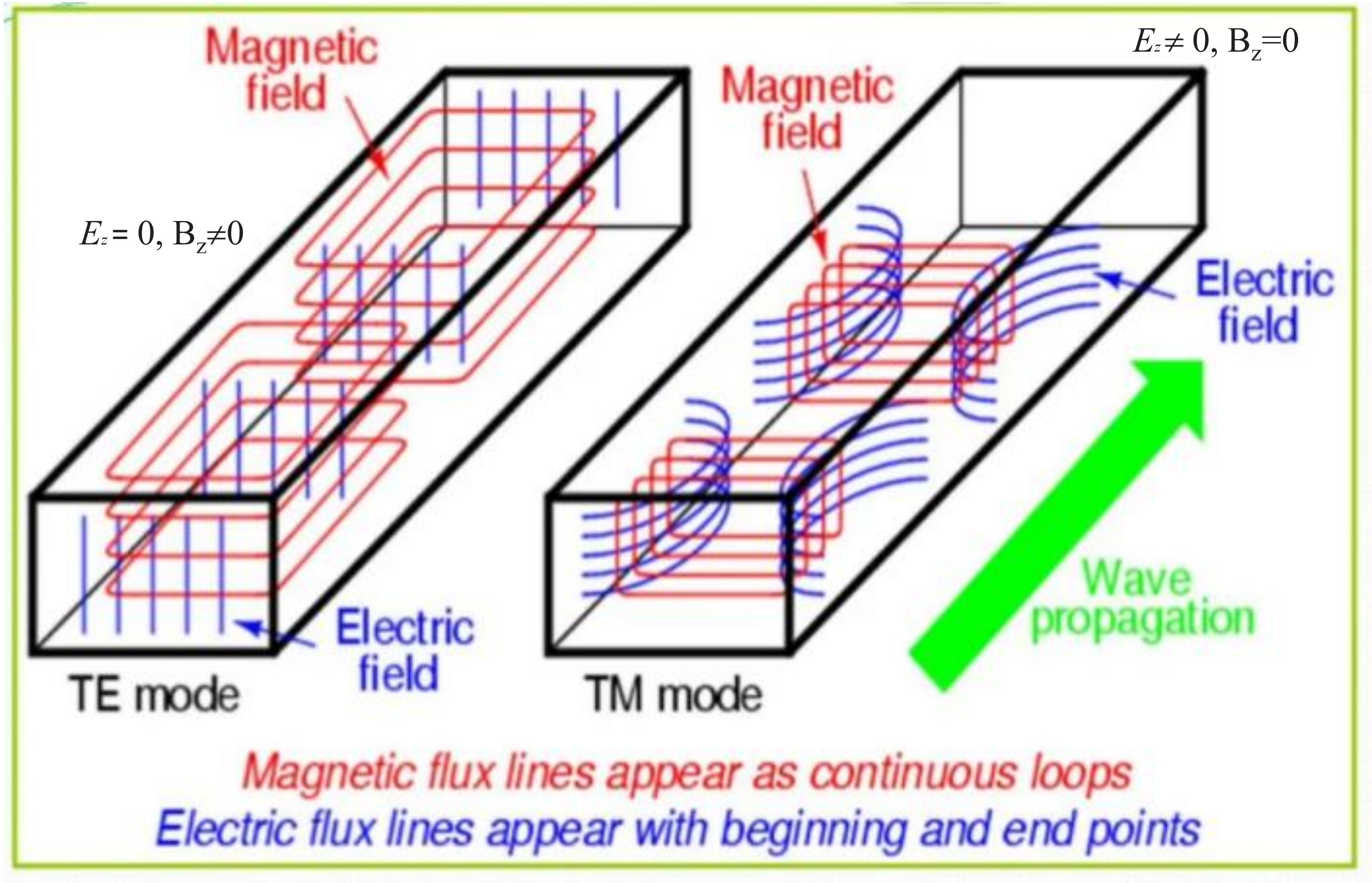
Case 3 Transverse electric and magnetic waves

Both $\widetilde{E_{0z}} = \widetilde{B_{0z}} = 0$

→no wave propagation, TEM Waves

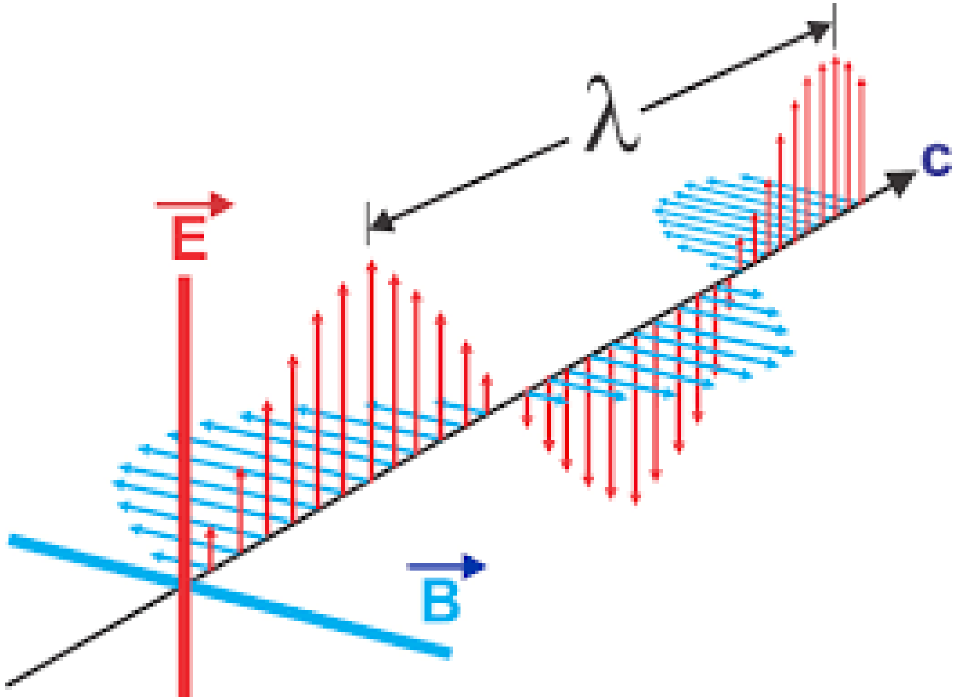
Hence, transverse electric and magnetic waves cannot propagate in hollow waveguides.

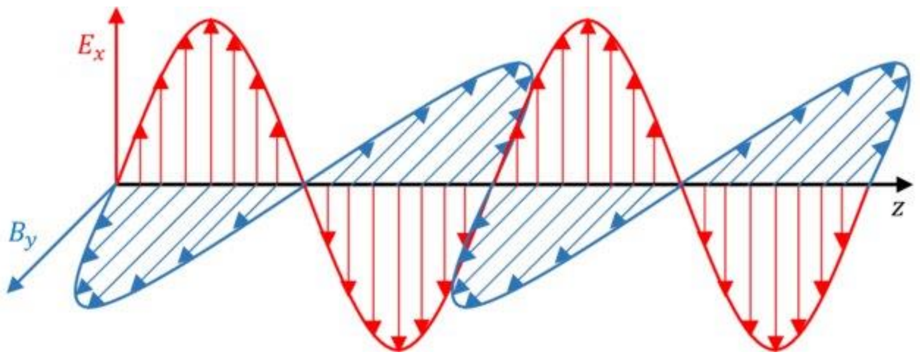
Basic features of waveguide



Thanks

EM Wave & Waveguides
Part-4





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Background (Already discussed)

$$\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\left(\frac{\omega}{c}\right)^2-k_z{}^2\right]\widetilde{E_{0z}}=0 \ldots\ldots\ldots\ldots\ (24)$$

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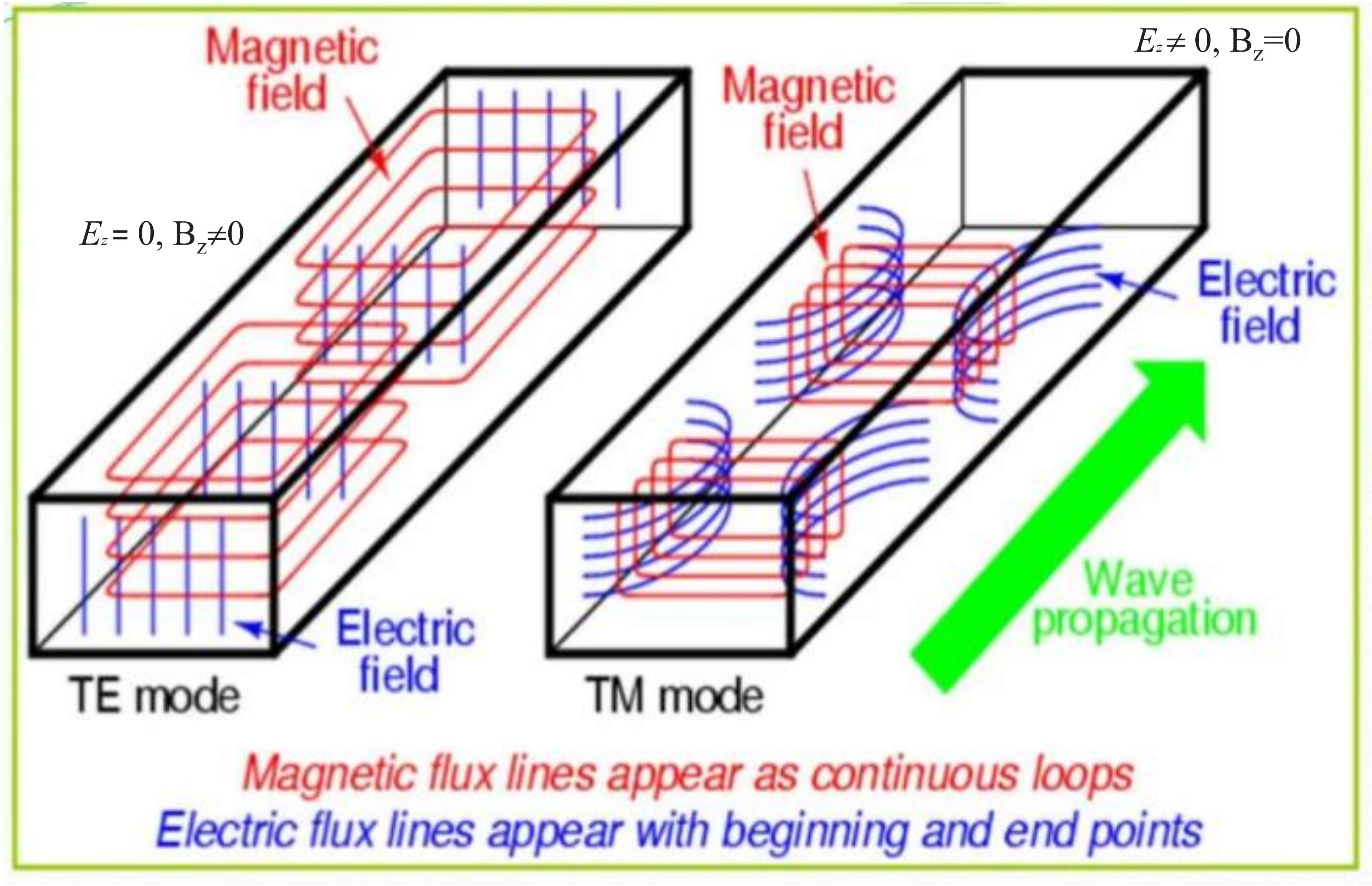
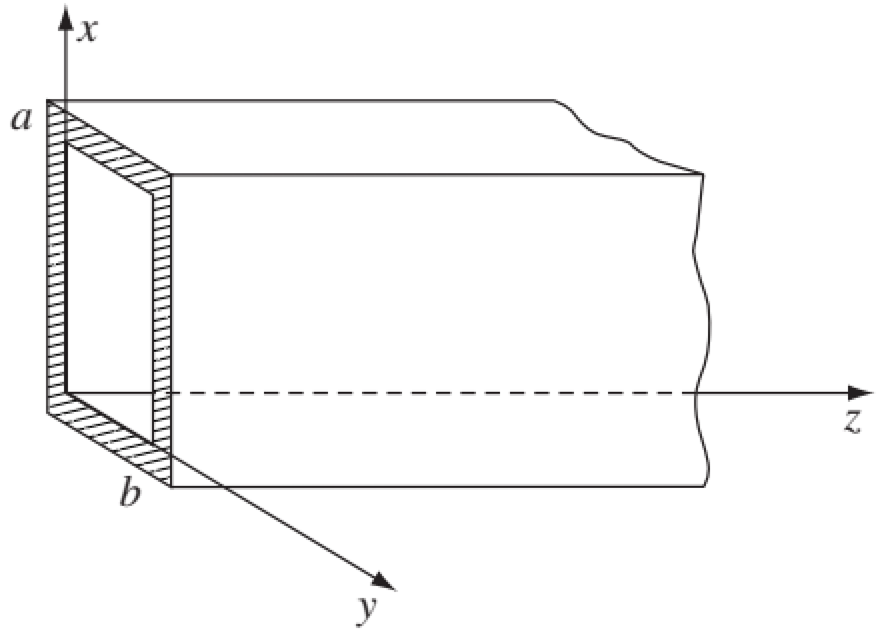


TE Waves in a Rectangular Wave Guide

Consider a wave guide of rectangular shape, with height a and width b, and we are interested in the propagation of TE waves.



$E_z = 0$, we call these **TE** ("transverse electric")

$$B_z(x, y) = X(x)Y(y), \qquad (1)$$

$$Y\frac{d^2X}{dx^2} + X\frac{d^2Y}{dy^2} + \left[(\omega/c)^2 - k^2\right]XY = 0.$$

Divide by XY , and note that the x- and y-dependent terms must be constants:

$$\text{(i)} \quad \frac{1}{X}\frac{d^2X}{dx^2} = -k_x^2, \quad \text{(ii)} \quad \frac{1}{Y}\frac{d^2Y}{dy^2} = -k_y^2,$$

With $$-k_x^2 - k_y^2 + (\omega/c)^2 - k^2 = 0. \qquad (2)$$

The general solution

$$X(x) = A\sin(k_x x) + B\cos(k_x x). \qquad (3)$$

But the boundary conditions require that

for TE waves $E_z = 0$ everywhere.

Subjected to condition $B^{\perp} = 0\,(B_x = B_y = 0)$

This implies that

$$\frac{\partial B_z}{\partial x} = \frac{\partial B_z}{\partial y} = 0$$

Or

$$\left.\frac{\partial B_z}{\partial n}\right|_s = 0$$

where **n** is the normal to the surface

dX/dx—vanishes at $x = 0$ and $x = a$. So $A = 0$, and

$$k_x = m\pi/a, \quad (m = 0, 1, 2, \ldots). \qquad (5)$$

The same goes for Y, with

$$k_y = n\pi/b, \quad (n = 0, 1, 2, \ldots), \qquad (6)$$

and we conclude that

$$B_z = B_0 \cos(m\pi x/a)\cos(n\pi y/b).$$

This solution is called the TE_{mn} mode. (The first index is conventionally associated with the *larger* dimension, so we assume $a \geq b$. By the way, at least *one* of the indices must be nonzero—see Griffiths Book)

The wave number (k) is obtained by putting Eqs. (5) and (6) into Eq. (2),

$$k = \sqrt{(\omega/c)^2 - \pi^2[(m/a)^2 + (n/b)^2]}.$$

If

$$\boxed{\omega < c\pi\sqrt{(m/a)^2 + (n/b)^2} \equiv \omega_{mn},}$$

the wave number is imaginary, and instead of a traveling wave we have exponentially attenuated fields. For this reason, ω_{mn} is called the **cutoff frequency** for the mode in question. The *lowest* cutoff frequency for a given wave guide occurs for the mode TE_{10}:

$$\omega_{10} = c\pi/a. \qquad \text{Or} \quad \omega_{1,0} = \frac{\pi}{\sqrt{\mu\epsilon}\, a}$$

Frequencies less than this will not propagate at all.

The wave number can be written more simply in terms of the cutoff frequency:

$$\boxed{k = \frac{1}{c}\sqrt{\omega^2 - \omega_{mn}^2}.}$$

The wave velocity (is greater than c)

$$\boxed{v = \frac{\omega}{k} = \frac{c}{\sqrt{1-(\omega_{mn}/\omega)^2}},}$$

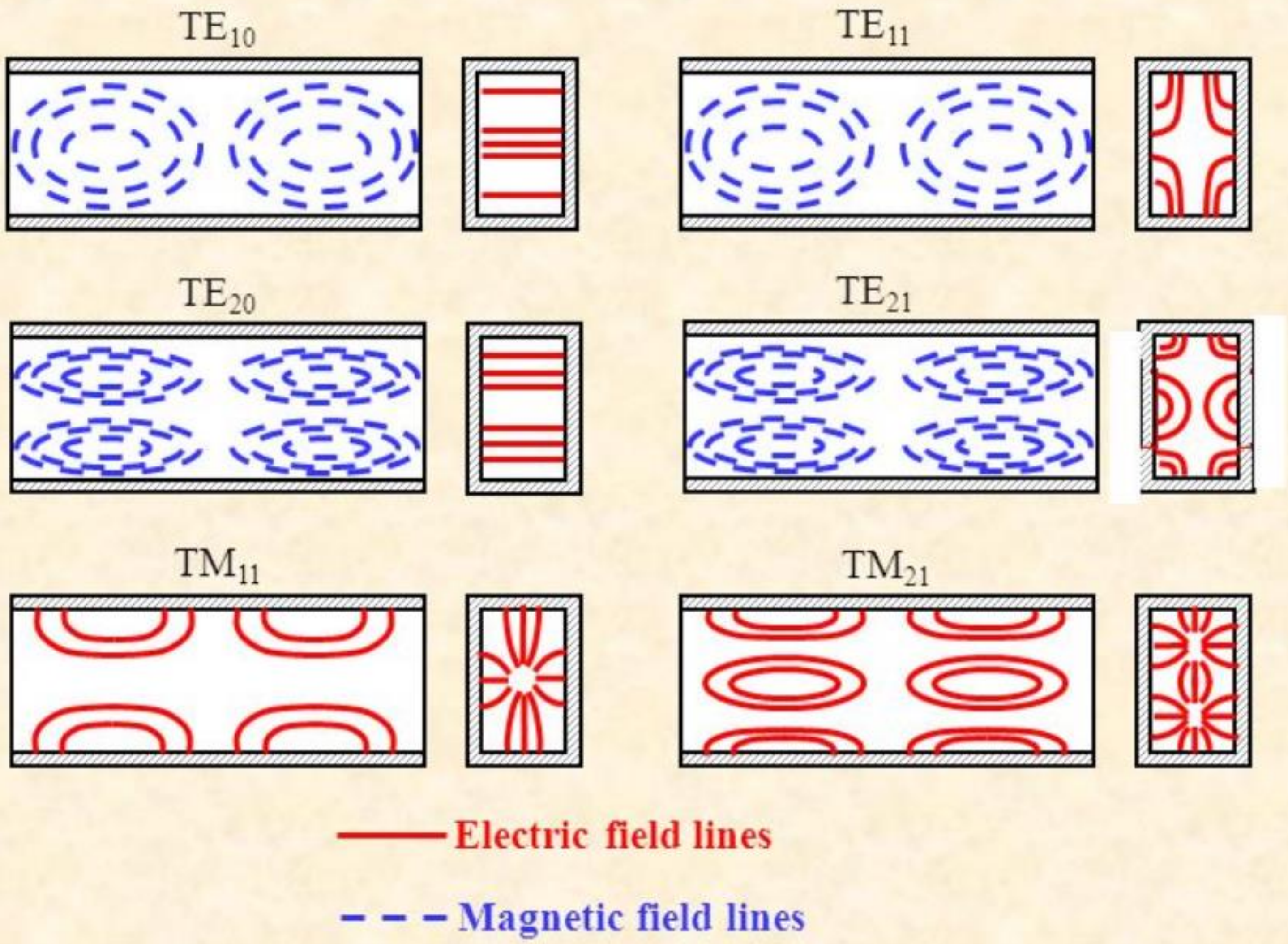
The energy carried by the wave travels at the *group* velocity

$$\boxed{v_g = \frac{1}{dk/d\omega} = c\sqrt{1-(\omega_{mn}/\omega)^2} < c.}$$

The explicit fields for the mode denoted by $TE_{1,0}$ are:

$$\boxed{\begin{aligned} H_z &= H_0 \cos\left(\frac{\pi x}{a}\right) e^{ikz-i\omega t} \\ H_x &= -\frac{ika}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{ikz-i\omega t} \\ E_y &= i\,\frac{\omega a \mu}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{ikz-i\omega t} \end{aligned}}$$

The modes with higher orders



Key features of waveguide

- Waveguides are used to carry energy between pieces of equipment or over longer distances to carry transmitted power to an antenna or microwave signals from an antenna to receiver.
- Waveguides are made from copper, aluminum. The metals are extruded into long rectangular or circular pipes.
- An electromagentic energy to be carried by a waveguide is injected in to one end of the waveguide.
- The electric and magnetic fields associated with the signal bounce off the inside walls back and forth as it progresses down the waveguide.


Rectangular waveguide

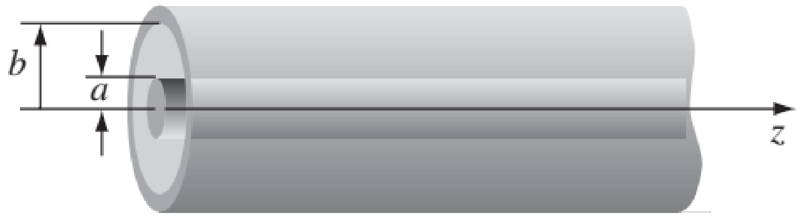

Waveguide to coax adapter


Waveguide bends


E-tee

The Coaxial Transmission Line

A hollow wave guide cannot support TEM waves. But a coaxial transmission line, consisting of a long straight wire of radius a, surrounded by a cylindrical conducting sheath of radius b, *does* admit modes with $E_z = 0$ and $B_z = 0$.



In this case Maxwell's equations yield

$$k = \omega/c$$

(so the waves travel at speed c, and are nondispersive),

$$cB_y = E_x \quad \text{and} \quad cB_x = -E_y$$

(**E** and **B** are mutually perpendicular), and (together with $\nabla \cdot \mathbf{E} = 0$, $\nabla \cdot \mathbf{B} = 0$)

$$\left.\begin{aligned} \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = 0, \quad \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0, \\ \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} = 0, \quad \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = 0. \end{aligned}\right\} \quad (1)$$

These are the equations of *electrostatics* and *magnetostatics*, for empty space, in two dimensions.

Hence the solution with cylindrical symmetry can be borrowed directly from the case of an infinite line charge and an infinite straight current, respectively.

$$\mathbf{E}_0(s, \phi) = \frac{A}{s}\,\hat{\mathbf{s}}, \qquad \mathbf{B}_0(s, \phi) = \frac{A}{cs}\,\hat{\boldsymbol{\phi}},$$

See Griffiths Book

for some constant A. Substituting these into

$$\left.\begin{aligned} \text{(i)}\;\; \tilde{\mathbf{E}}(x, y, z, t) &= \tilde{\mathbf{E}}_0(x, y)e^{i(kz-\omega t)}, \\ \text{(ii)}\;\; \tilde{\mathbf{B}}(x, y, z, t) &= \tilde{\mathbf{B}}_0(x, y)e^{i(kz-\omega t)}. \end{aligned}\right\}$$

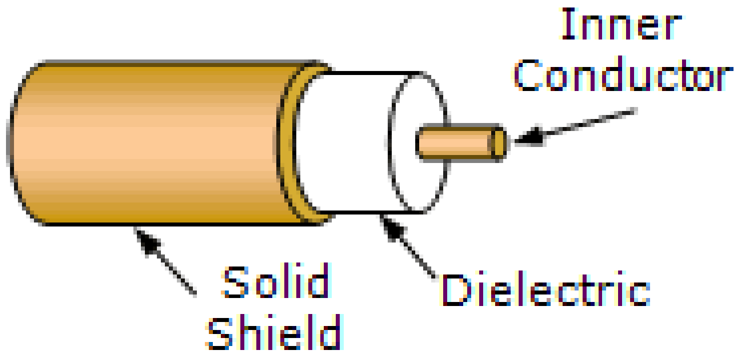
and taking the real part:

$$\left.\begin{aligned} \mathbf{E}(s, \phi, z, t) &= \frac{A\cos(kz-\omega t)}{s}\,\hat{\mathbf{s}}, \\ \mathbf{B}(s, \phi, z, t) &= \frac{A\cos(kz-\omega t)}{cs}\,\hat{\boldsymbol{\phi}}. \end{aligned}\right\}$$

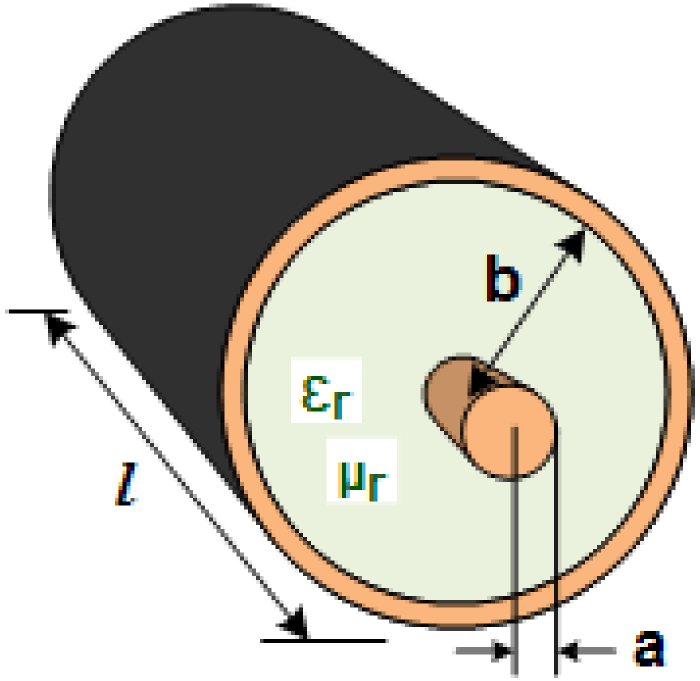
The coaxial cable is one of the common transmission example -cable TV. A typical experiment literally uses miles of transmission lines. The transmission line consists of an inner conductor – a wire – and an outer conductor – usually a metal braided jacket.

The Coaxial Transmission Line:

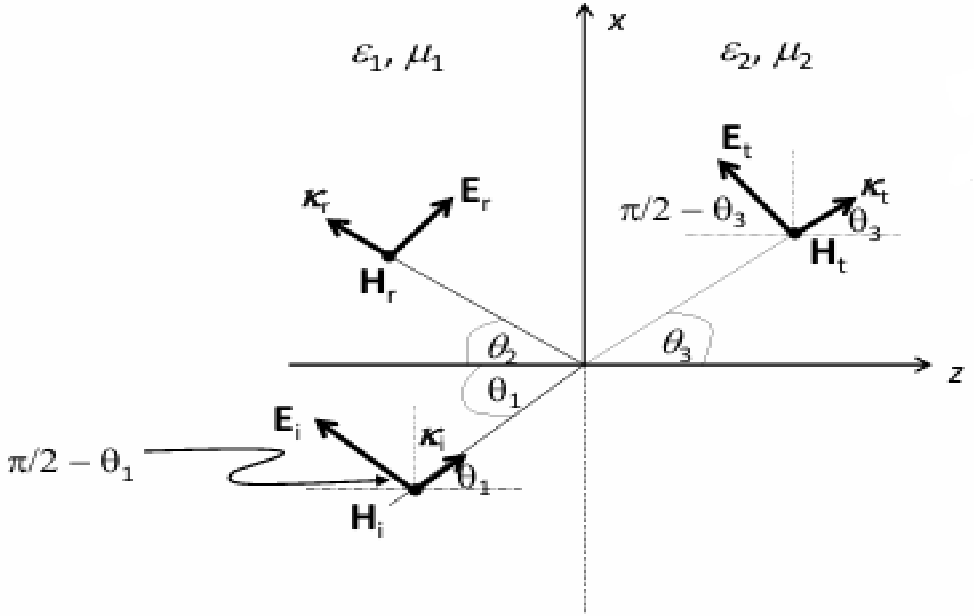
Geometrical Information



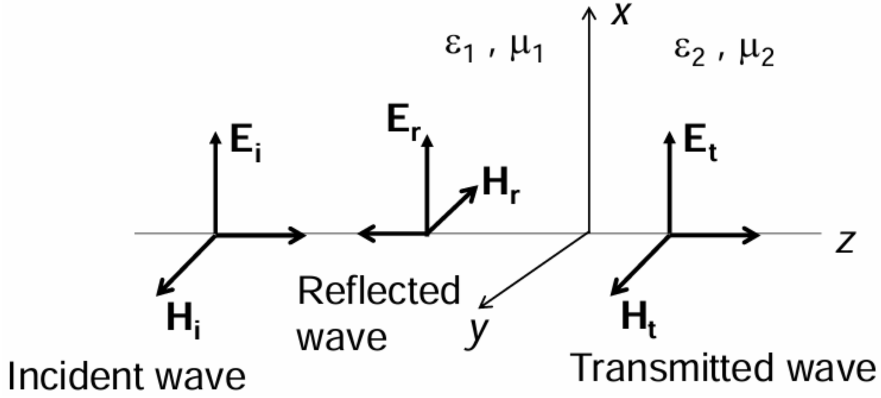
a = outside radius of inner conductor (inches)
b = inside radius of outer conductor (inches)
c = speed of light in a vacuum
$\boldsymbol{\varepsilon}$ = dielectric constant = $\boldsymbol{\varepsilon}_0 * \boldsymbol{\varepsilon}_r$
$\boldsymbol{\varepsilon}_0$ = permittivity of free space
$\boldsymbol{\varepsilon}_r$ = relative dielectric constant
$\boldsymbol{\mu}_r$ = relative permeability



Reflection and Transmission at the boundary between two dielectrics for oblique incidence



Simple case



$$\overrightarrow{E_i} = \hat{x}\, E_{10}\, e^{i(\omega t - k_1 z)}$$
$$\overrightarrow{E_r} = \hat{x}\, E_{20}\, e^{i(\omega t + k_1 z)}$$
$$\overrightarrow{E_t} = \hat{x}\, E_{30}\, e^{i(\omega t - k_2 z)}$$

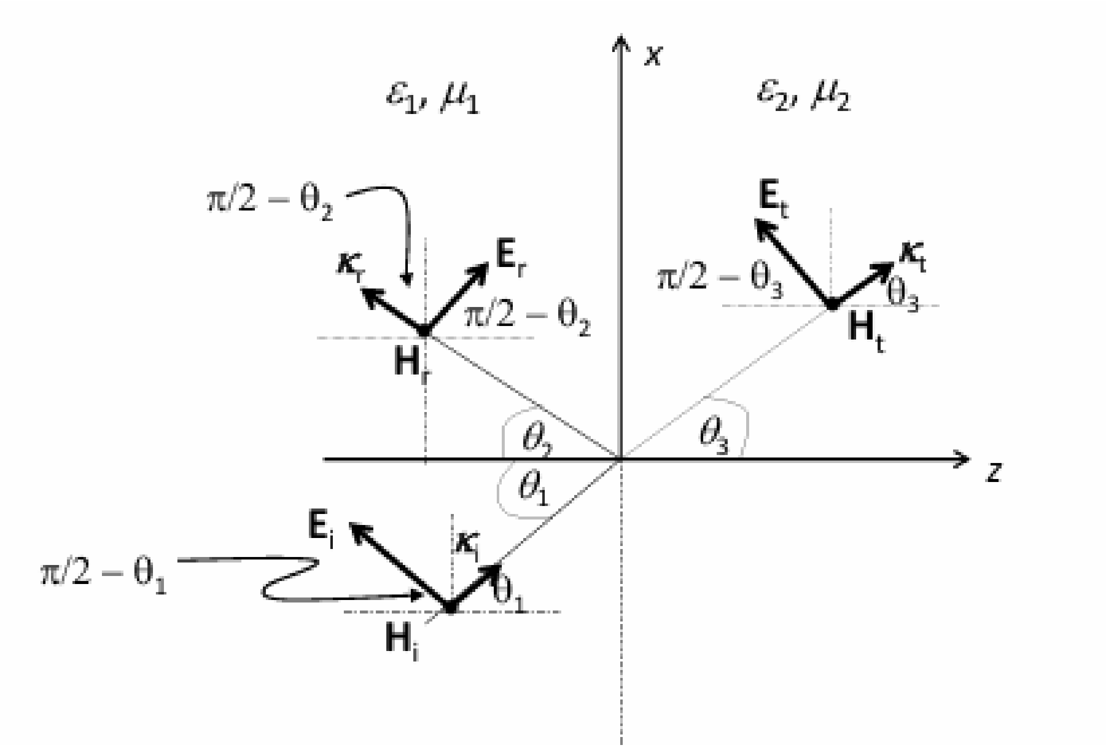
$$k_1 = \omega\sqrt{\mu_1 \varepsilon_1} \qquad \& \quad k_2 = \omega\sqrt{\mu_2 \varepsilon_2}$$

$$k_i = \omega\sqrt{\varepsilon_1 \mu_1}\,(\hat{x}\sin\theta_1 + \hat{z}\cos\theta_1)$$
$$k_t = \omega\sqrt{\varepsilon_2 \mu_2}\,(\hat{x}\sin\theta_3 + \hat{z}\cos\theta_3)$$

$$\overrightarrow{E_i} = \overrightarrow{E_{1,0}}\, e^{i(\omega t - \overrightarrow{k_i}.\vec{r})}$$
$$\overrightarrow{E_t} = \overrightarrow{E_{3,0}}\, e^{i(\omega t - \overrightarrow{k_t}.\vec{r})}$$

where,

$$\overrightarrow{E_{1,0}} = E_{1,0}\,(\hat{x}\ \cos\theta_1 - \hat{z}\sin\theta_1)$$
$$\overrightarrow{E_{3,0}} = E_{3,0}\,(\hat{x}\ \cos\theta_3 - \hat{z}\sin\theta_3)$$



$$\overrightarrow{E_r} = \overrightarrow{E_{2,0}}\, e^{i(\omega t - \overrightarrow{k_t}\,.\vec{r})}$$

where

$$\overrightarrow{E_{2,0}} = E_{2,0}\,(\hat{x}\cos\theta_2 + \hat{z}\sin\theta_2)\ ;$$

$$k_r = \omega\sqrt{\varepsilon_1\mu_1}\,(\hat{x}\sin\theta_2 - \hat{z}\cos\theta_2)$$

From the boundary conditions :

Tangential component ⇒ x-component of **E** at $z = 0$ is continuous, it can be shown that

$$(\ )e^{i(\omega t - \overrightarrow{k_i}\,.\,\overrightarrow{r})} + (\ )e^{i\left(\omega t - \overrightarrow{k_r}\,.\vec{r}\,\right)} = (\ )e^{i(\omega t - \overrightarrow{k_t}.\vec{r}\,)}$$

$$k_i \sin\theta_1 = k_r \sin\theta_2 = k_t \sin\theta_3$$

$\theta_1 = \theta_2$: Law of reflection

Also,

$n_1 k_0 \sin\theta_1 = n_2 k_0 \sin\theta_3$ $k_0=\omega/c$

$\Rightarrow n_1 \sin\theta_1 = n_2 \sin\theta_3$: Snell's Law

From the boundary conditions for the tangential and normal components of E, H :

$$E_{1,0} \cos\theta_1 + E_{2,0} \cos\theta_1 = E_{3,0} \cos\theta_3$$

$$-\varepsilon_1 \sin\theta_1 E_{1,0} + \varepsilon_1 \sin\theta_1 E_{2,0} = -\varepsilon_2 \sin\theta_3 E_{3,0}$$

After some algebraic manipulations :

$$T = \frac{E_{3,0}}{E_{1,0}} = \frac{2\,\varepsilon_1 \sin\theta_1 \cos\theta_1}{\varepsilon_2 \sin\theta_3 \cos\theta_1 + \varepsilon_2 \sin\theta_1 \cos\theta_3}$$: transmitted field

$$R = \frac{E_{2,0}}{E_{1,0}} = \frac{\varepsilon_1 \sin\theta_1 \cos\theta_3 - \sin\theta_3 \cos\theta_1}{\varepsilon_2 \sin\theta_3 \cos\theta_1 + \varepsilon_1 \sin\theta_1 \cos\theta_3}$$: reflected field

After further algebraic manipulations :

$$\frac{E_{2,0}}{E_{1,0}} = -\frac{\tan(\theta_1 - \theta_3)}{\tan(\theta_1 + \theta_3)}$$

$$\frac{E_{2,0}}{E_{1,0}} = -\frac{\tan(\theta_1 - \theta_3)}{\tan(\theta_1 + \theta_3)}$$

Observe that

1. For $\theta_1 = \theta_3 \Rightarrow \tan(\theta_1 - \theta_3) = 0 \Rightarrow No\ reflection$!

   Possible if $n_1 = n_2 \Rightarrow$ two media are same i.e optically indistinguishable

2. For $\theta_1 + \theta_3 = \pi/2 \Rightarrow \tan(\theta_1 - \theta_3) = \infty$

$\Rightarrow$ No reflection !

From Snell's Law we have

$$n_1 \sin\theta_1 = n_2 \sin\theta_3$$

$\Rightarrow For\ \theta_3 = \frac{\pi}{2} - \theta_1$

$$n_1 \sin\theta_{1p} = n_2 \cos\theta_{1p}$$

$\Rightarrow \tan\theta_{1p} = \frac{n_2}{n_1}$: Brewster law

↓

Brewster angle

Brewster's law states that when unpolarized light passes through a transparent substance at a specific angle, the light that is reflected is perfectly polarized

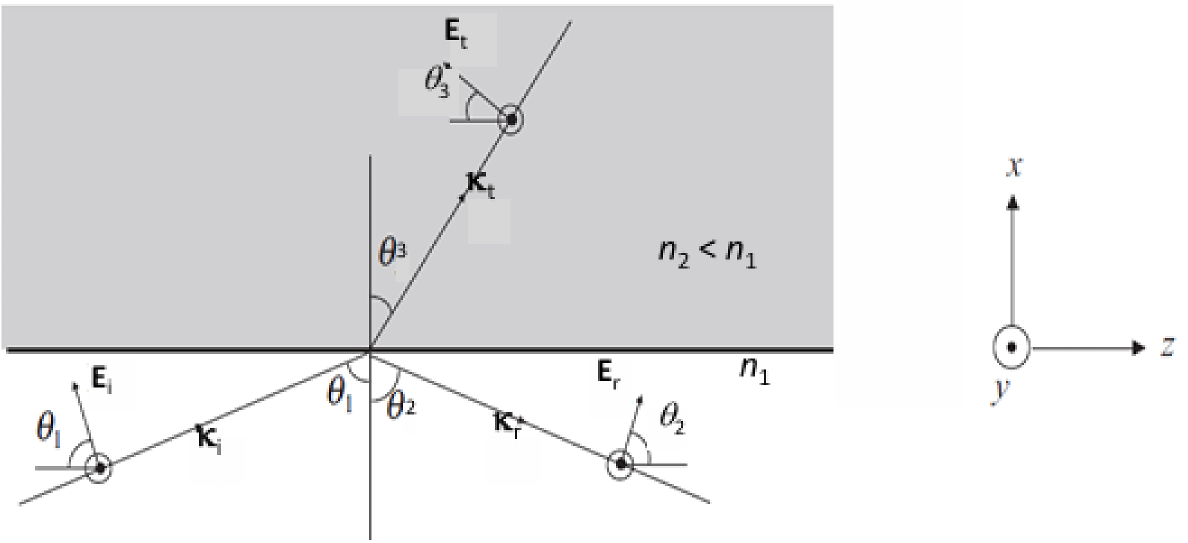
For glass and air :

$$n_2 = 1.5\,, n_1 = 1 \;\Rightarrow\; \theta_p = \tan^{-1}(1.5) = 56.3^0$$

Reflection and transmission of an em wave at an interference that separate two dielectrics $\varepsilon_{1,2}$

$$n_{1,2} = \sqrt{\varepsilon_{1,2}/\varepsilon_o}$$

Special Case : $n_2 < n_1$



From Snell's Law

$$n_1 \sin\theta_1 = n_2 \sin\theta_3 \;\; \Rightarrow \; \theta_3 > \theta_1$$

$$\sin\theta_3 = \frac{n_1}{n_2}\sin\theta_1 = \sqrt{(\varepsilon_1/\varepsilon_o)(\varepsilon_2/\varepsilon_o)}\;\sin\theta_1 \Rightarrow \sin\theta_3 = \sqrt{\varepsilon_1/\varepsilon_2}\sin\theta_1$$

Consider an angle $\theta_1 = \theta_c = critical\ angle$ for which $\sin\theta_3 = 1$

$$\Rightarrow \theta_c = \sin^{-1}\left({}^{n_2}/_{n_1}\right) = \sin^{-1}\sqrt{\varepsilon_2/\varepsilon_1}$$

$\Rightarrow$ For $\theta_1 > \theta_c$,

$\Rightarrow \sin\theta_3 > 1$!!

$\Rightarrow$ For $\theta_1 > \theta_c$,

The incident wave returns to medium 1 of R.I. n_1

$\Rightarrow$ Total internal reflection (TIR)

Thanks